Online Resource 1 Supplementary information

SCLARA: linking paternal geography to dated archaeological landscapes reveals distinct lineage associations in Sweden

Archaeological and Anthropological Sciences

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This supplement documents the sources, fixed archaeological measurements, complete set of comparisons and interpretation of the Swedish results. Current four-period and later-profile results are distinguished from earlier posthole–hearth analyses used to evaluate source-pool sensitivity and estimator performance. Posterior intervals, inverse-Hessian intervals and empirical sensitivity ranges have different meanings throughout.

Geographic reference data include GeoNames (2026) populated places and the Institute for Language and Folklore (2026) parish and province references. Modern country outlines use Natural Earth (undated); regional restrictions use Eurostat (2024) GISCO NUTS boundaries. © EuroGeographics for the administrative boundaries. These references support locality reconciliation, geographical screening and display; they do not reconstruct ancient borders. Dataset versions, acquisition locators and checksums are recorded in SOURCES.tsv.

Analytical development and scope

The initial Swedish screen covered thirteen period panels and 74 feature classes. Its finite representation grid produced 1,664 axes, 1,352 distinct fitted designs and 9,984 lineage contrasts. Exact repetitions of a genetic design are not independent tests. All screen-wide Holm-adjusted values were 1. Twelve profiles received focused historical appraisal with their results known. A selected comparison is therefore not an independently confirmed finding simply because its archaeological decomposition omitted lineage labels.

The subsequent posthole–hearth comparison held a feature contrast constant across four discontinuous windows. A broader prehistoric extension grouped features into seven families and constructed one common Late Neolithic/Bronze Age basis. The present four-period model keeps that basis fixed while adding Roman and Migration records, applying source corrections and retaining all three components and seven lineage contrasts. The two preferred Viking/medieval profiles and their companion representations are also preserved with their original transformations. They were selected for archaeological information before their updated effects were fitted. No new all-period axis screen accompanies these follow-ups.

Historical coherence was originally recorded on a 0–3 rubric: contradicted or uninterpretable after review; provisional source-linked dated context; phase-appropriate functional interpretation supported on both poles; or evidence connecting that interpretation to competing historical predictions. Unreviewed cases remained unscored. These levels summarize documented appraisal rather than probabilities or validated quality scores. Later source review can refine an interpretation without making it a demonstrated origin mechanism. A credible unresolved contrast can be retained for the historical distinction it addresses.

Paternal sources and units

The frozen source union comprises 6,785 records from Visual DNA, the FamilyTreeDNA Sweden project and Boström's southern Norrbotten compilation (Kinnunen undated; FamilyTreeDNA 2026; Boström undated). Source acquisition on 2–3 November 2026 fixes the observed versions; ancestral reference dates are retained separately. The geographical layer represents reported paternal ancestral localities. Exact identifiers and documented family links support reconciliation. Place-name similarity and broad branch compatibility alone do not establish identity. Screening retains 3,633 geographically and phylogenetically

eligible records. Among observed nested labels, the deepest represented labels are retained and 1,986 upstream-labelled records are withheld. This yields 1,647 selected records and 1,340 non-nested paths, reconciled into 1,339 units at 1,404 distinct positions.

The common-period model excludes four whole units because terrain is unavailable at one or more of their reported localities. Its 1,335 units contain 1,641 source records and 1,594 unit-locality rows using 1,399 distinct positions. The full measurement arrays retain 1,404 positions; five are unused by these fits. The later models retain all 1,339 units with their original missing-terrain controls. Broader dependence blocks number 246 in both cohorts. A branch unit is not an independently verified family, and its several origins need not refer to the same person. Each distinct retained origin receives equal weight within its unit.

Table S1 gives the common-period cohort by lineage. Across this cohort, 979 units have an exact FamilyTreeDNA table link; 978 of these have a nonempty ancestral statement in that table. There are 743 units (55.7%) with at least one reported ancestral birth year, 163 with an uncertain or conflicted date and 30 with an exact curated-compilation link; categories overlap. The 846 date-bearing source records span reported birth years 1169–1960, with median 1724 and interquartile range 1665–1786. Of these records, 672 (79.4%) report births before 1800 and 816 (96.5%) before 1900. These are record-level descriptive summaries; repeated ancestral information does not add likelihood weight to a reconciled unit. The geographical reference has variable genealogical depth: a reported birth year does not uniformly establish residence at the mapped point in that year. Empirical positional-error radii remain unavailable.

Table S1 Provenance and dependence of the common-period paternal cohort

Lineage	Units	Blocks	Records	FTDNA linked	Birth year	Curated link	Multiple localities
I1-L22	236	47	290	170	138	5	44
I1-Z58	206	35	257	153	116	3	44
I1-other	95	20	116	67	57	2	15
R1a-Z284	154	21	195	119	91	5	32
R1b-U106	157	37	185	117	96	1	17
R1b-P312	144	23	171	119	75	5	23
N	125	22	156	67	48	6	23
Other	218	41	271	167	122	3	42

There are 240 multi-origin units; their maximum internal separation exceeds 100 km in 75 units and 200 km in 50, reaching approximately 1,214 km. These distances describe heterogeneity of represented origins, not measured coordinate errors. Testing resolution also changes representation. The two fixed branch cuts, capped at two and five observed descendant labels, recover additional upstream-labelled records and group the units more broadly. Their construction and all four earlier posterior comparisons are reported below; they do not recover unobserved pedigrees.

Archaeological sources and chronology

Kulturmiljöregistret (KMR; Riksantikvarieämbetet 2026) supplies native point, line and polygon geometry, registered attributes and source uncertainty where available. The retained native subset contains 6,107 sites, 7,424 geometries, 6,378 components and 4,392 properties. Correcting the Örtofta link adds the appropriate site and polygon (Bolander 2016) while retaining the originally linked record. The full Strucke (2026) version 2 table contains 30,296 assertions, despite a repository-description count of 30,301.

Original row text, laboratory identifiers, context descriptions and disagreements remain accessible. Unmatched or withheld assertions are not erased.

Representative points are projected centroids for the appropriate geometry, not the true nearest boundary or a surveyed sample position. Shared archaeological positions receive shared displacements. The displacement distance is the retained radius multiplied by the square root of a uniform draw on [0, 1], applied along a great circle with a uniform bearing. Existing positions preserve their radii and random streams; new shared positions use the largest available declared source uncertainty. Unknown uncertainty remains unknown; a nominal undisplaced position under the inherited convention does not imply an empirical zero-error estimate. Modern boundaries and incomplete foreign-site coverage remain limitations near the national edge.

Annual radiocarbon weights combine laboratory error with the IntCal20 mean and variance (Reimer et al. 2020). For a reported age and error, the likelihood is a normal density evaluated against each annual calibration-curve value, with laboratory and curve variances added. Calendar years from 0 to 55,000 cal BP receive a uniform prior; years for which the reported radiocarbon age differs from the curve mean by more than eight combined standard deviations receive zero weight before normalization. Identical normalized laboratory determinations share draws. Conflicting reported age/error combinations are withheld or explicitly adjudicated from the original evidence. The procedure does not estimate covariance among different determinations or material-to-event offsets.

For the common profiles, one material-date draw is paired with one archaeological-position draw in each of 64 realizations. Period bounds are half-open and fixed: Late Neolithic [−2300, −1700), Bronze Age [−1700, −500), Roman Iron Age [1, 400), Migration [400, 550). Negative numerical years label BCE in this implementation; the analysis does not infer precision at a historical year-zero boundary. A draw outside all windows contributes to none. Assertions are collapsed at site, family and period before nearest-feature measurement. Support uses a site probability threshold of 0.1 and at least ten sites across three counties, rather than complete-envelope containment.

Table S2 Qualifying distinct sites and counties for every family and target period

Family	Late Neolithic	Bronze Age	Roman	Migration
Building traces	130 / 15	321 / 20	460 / 19	324 / 19
Combustion	120 / 18	1086 / 20	1050 / 21	715 / 21
Pits and deposits	88 / 16	282 / 18	217 / 17	116 / 15
Heated stone	16 / 11	118 / 15	36 / 12	14 / 7
Clearance/cultivation	13 / 8	81 / 15	95 / 14	70 / 13
Burial	22 / 6	112 / 14	113 / 15	85 / 14
Water	12 / 5	36 / 7	53 / 9	20 / 7

The current common-period pool has 14,566 assertions at 3,640 sites. All 28 target family-period support checks pass. The initially considered 2800–2300 BCE panel failed comparable support and was not fitted. A preselected phase-omission scenario removes nine reviewed older-ground, underlying or probably reworked activity determinations; Late Neolithic clearance then has eight qualifying sites and fails the primary rule. The fitted coefficients describe sensitivity to source selection; material dates remain distinct from securely established activity dates.

The later profiles use their original complete-containment rule in AD 750–1050 and AD 1050–1520. They retain sixteen Viking and twenty-six medieval feature classes, with at least five positions and three source

groups per included feature. The reviewed panel contains 441 Viking and 1,055 medieval assertion links. Seven link changes comprise the withheld Selånger posthole, three Säby site corrections and three inherited Örtofta corrections. The feature definitions, transform and role of each preferred or companion representation remain fixed.

Source qualification and archaeological meaning

The clearance review covered fifteen sites that initially qualified and three additional sites that contributed substantially to the measurement. Fourteen received substantive original-context appraisal. The available sources did not permit a full assessment of Visby, Säby, Finja and Hogstad; these contexts remain unresolved. The review distinguishes cultivation, grazing clearance and undetermined clearance, and records whether the dated material could precede the inferred activity. It also asks whether buildings, crops or environmental evidence independently support the interpretation.

At Mannaberg (Nordin 2023), horizon A1462, sample PK1000944 and macrofossil sample PM1994 link the source assertion to the excavated context. Ash charcoal was dated to 3449 ± 30 BP. The horizon contains one unidentified cereal grain and charcoal/brushwood; its interpretation as cultivated soil is supported by the report, but the grain itself was not dated. Earlier pottery and residual material prevent a uniquely dated cultivation episode. Kölbygärde (Ring et al. 2001) contains older charcoal beneath later land-use deposits; Kråkhult (Hellgren 2023) favours grazing improvement and retains a context-number discrepancy within the same registered clearance site. These examples preserve the archaeological content without making every clearance feature contemporary cultivation.

The Öjaby check (Emilsson et al. 2020) examined all 39 determinations on the original laboratory certificates. Of the 35 determinations already in the catalogue, all had mismatched context identifiers; twenty also had different feature assignments and two had different site assignments. These discrepancies, together with one age and one error parameter, were corrected in the analytical records. Four absent pollen determinations are not introduced into the feature model. Nine previous eligibility barriers are removed where their basis was mismatched metadata. The older ground horizon beneath later cairn fill remains distinct from direct cultivation evidence. The Räppe age conflict remains unresolved. Rödje and Glasskärvan corrections follow Färjare (2021) and Petersson and Zetterlund (2021), respectively. KMR identities and the Marieberg report establish the specified Säby crosswalk (Balknäs 2020); KMR resolves the specified Stoby identity. Bådstorp context links follow Seiler and Sjölin (2022). These identity repairs do not supply the missing original Säby or Stoby activity contexts. Original values and report disagreements remain attached to each decision.

The later seven-report review covers Färila charcoal production, Harbo-Smedsbo iron-production waste, the Fyrislund hearth, Selånger contexts, Leksand tar production, Nifsarp charcoal production and Uppsala brick production. It checks both poles of the selected profiles. Färila lacks local iron-production remains around the reviewed charcoal pit; the adjacent charcoal pit at Leksand is much later than its tar context; Nifsarp has reversed radiocarbon/stratigraphic ordering; and Uppsala's oven is a brick kiln. The medieval positive pole therefore cannot be labelled uniformly domestic or urban.

At Selånger (Strucke and Bäck 2003), the dating table, context catalogue and narrative disagree about three dated-feature links. The house discussion also rejects the cereal dates as a satisfactory chronology relative to secure brick evidence. Assertions C14:14111, C14:14112 and C14:14113 are withheld, with two prior holds preserved. C14:14037 remains clearance evidence with its context-number conflict recorded. Adjudication preserves conflicts between the report, dating table and context catalogue; no replacement laboratory age is established by those conflicting sources. These source reviews are targeted appraisals, not a random validation sample of all Swedish sites.

Frozen transformations and model units

For a period, family, realization and paternal locality, proximity P is exp(−d/50), where d is the nearest retained representative-point distance in kilometres. The maximum site proximity is taken within each realization before averaging. A realization with no eligible site for a family in the admitted source pool gives zero proximity. With seven family proximities, relative measurement subtracts their within-period mean.

$$P_{\mathrm{tfr}}(x) = \exp(\frac{-d_{\mathrm{tfr}}(x)}{50})$$

The basis was fitted to the 64-realization mean relative proximities, stacking the two period-specific locality matrices with equal total weight per paternal unit and period. The three leading eigenvectors are oriented so that each largest absolute loading is positive. Scores multiply the standardized relative-feature matrix by these eigenvectors and divide componentwise by the square root of their eigenvalues. Reference explained fractions are 23.55%, 20.31% and 17.60%. New periods and source corrections preserve the loadings, reference scales and orientation. All three components enter the common-period model jointly; Component 3 supplies the principal archaeological interpretation.

Writing Q for the mean relative-family measurement, Z for its standardized columns, V for the retained eigenvectors and Λ for their diagonal eigenvalue matrix gives the transformation below. Reference μ and σ, V and Λ are identical in every projected period.

$$S = ZV\Lambda^{-\frac{1}{2}}, Z_{\mathrm{tf}} = \frac{Q_{\mathrm{tf}} - \mu_f}{\sigma_f}$$

Table S3 Frozen reference centres, scales and eigenvector loadings

Family	Centre	Scale	PC1	PC2	PC3
Building traces	0.074080	0.124967	−0.505805	0.067952	0.366815
Combustion	0.184632	0.133909	−0.068132	−0.476722	−0.468793
Pits and deposits	0.081393	0.134569	−0.455577	−0.282006	−0.175181
Heated stone	−0.072415	0.175636	0.556939	−0.033900	−0.329875
Clearance/cultivation	−0.061896	0.177612	0.269066	−0.364427	0.680824
Burial	−0.044966	0.148647	−0.319953	0.438256	−0.201448
Water	−0.160828	0.164133	0.216846	0.602135	0.045508

The fixed basis represents 57.6% of Roman and 55.4% of Migration within-period standardized variation. The Roman mean scores remain within the reference ranges. Migration Component 3 scores fall outside the frozen Late Neolithic/Bronze Age reference range for only 0.075% of represented locality weight. These are transport diagnostics, not proof of equal archaeological opportunity. Proximity-equivalent coverage within 100 km is only 38.7% for Migration heated stone and 41.8% for water features. Relative to the original reference measurement, the unit-weighted RMS change in Late Neolithic Component 3 is 0.200 reference-score units after reported-context repairs and 0.455 when the specified phase omission is also applied.

The later preferred Viking score uses relative exponential proximities with correlation scaling and the second unrotated component of a two-component solution. Its positive feature–axis correlations emphasize charcoal pits and slag heaps, while postholes and hearths are negative. Its companion is the third varimax component of a three-component solution, with broader additive positive content. The preferred medieval

score uses relative exponential proximities, functional-block scaling and the second unrotated component of a two-component solution; its companion uses relative Gaussian proximities and a three-component varimax solution. All full loading matrices and score coefficients remain fixed. These four axes cannot be substituted for one another or for the common basis.

Table S4 Selected feature–axis correlations in the four later representations

Feature	Viking preferred	Viking companion	Medieval preferred	Medieval companion
charcoal pit	0.640	0.852	−0.352	−0.030
slag heap	0.505	0.771	−0.244	−0.047
posthole	−0.510	−0.119	0.484	0.595
hearth	−0.388	0.093	0.335	0.538
oven unspecified	0.112	0.576	0.602	0.861
well	−0.290	0.202	0.502	0.684
tar kiln	—	—	−0.464	−0.380
cultural layer	0.344	0.811	0.060	0.203
stone setting	0.289	0.733	−0.137	0.286
floor layer	−0.131	0.404	0.280	0.556

Likelihood, priors and numerical inference

For each represented unit, softmax probabilities are calculated at every retained origin. The probability assigned to its observed category is the equally weighted mean of those probabilities. The logarithm of that mean contributes once to the likelihood. The implementation retains eight outcome categories and a reference-category parameterization. A reference category is a computational convention; the prior is symmetric in pairwise category differences.

For n units with m_i origins each, y_i denotes the observed category and x_{ij} its jth locality profile. The unit-mixture likelihood is

$$\ell(\beta) = \sum_i \log[\sum_j \frac{1}{m_i} p(y_i|x_{ij},\beta)]$$

For each predictor, the seven nonreference coefficients have a multivariate normal prior with diagonal variance s^2 and off-diagonal covariance $s^2/2$. Every pairwise category difference consequently has standard deviation s. The values are 2 for the intercept and regional terms, 0.5 for nuisance controls and 1 for archaeological slopes. Optimization, posterior sampling and weighted sensitivities use the same prior precision. Conditional modes require finite solutions, positive local curvature and maximum absolute gradient below 10^{-5}.

The five common-period non-archaeological controls are linear geographic coordinates, log genetic-registry density, elevation and local relief. Density uses the 3,633 screened source rows in 25 km azimuthal-equidistant cells and is a recruitment proxy, not a population denominator. Terrain uses the retained unresampled HydroSHEDS crop (HydroSHEDS undated; Lehner et al. 2008); north of 60° N its coarser HYDRO1k basis limits the information implied by nominal grid spacing. Recording is the mean log(1 + count) of eligible distinct positions within 75 km. Each period also has a mean-family-proximity opportunity control. Frozen means and scales are used throughout matched sensitivities.

The common-period regional basis uses 200 km SWEREF 99 TM cells, under original and 100 km-shifted origins. Archaeological scores are centred within cells using frozen reference means. Quadratic specifications retain the linear geographic controls and add standardized squared projected northing, northing-by-easting and squared easting. The later profiles retain their original ten environmental/recruitment controls, including terrain missingness, plus recording and opportunity. They do not inherit the common-period model's regional intercepts or become matched temporal effects.

All composite I1 contrasts retain the frozen weights 237/539, 207/539 and 95/539 for L22, Z58 and other I1. The common cohort has 236, 206 and 95 units after terrain exclusions; those counts do not redefine the composite estimand. Weights remain fixed across omissions and matched comparisons. The seven retained contrasts include I1–P312, derived directly from aligned coefficients. Four periods plus six pairwise differences, three components and seven lineage contrasts give 210 contrasts per common-period fit, or 840 summaries across four national specifications. Later profiles retain seven contrasts each, including unresolved comparisons.

Four chains begin with 1,000 warmup and 1,000 retained draws each. Target acceptance is 0.95 and maximum tree depth 12. Convergence diagnostics follow Vehtari et al. (2021). Diagnostic failure alone permits one extension to 2,000 warmup and retained draws per chain, acceptance 0.99 and depth 14. Required R-hat is at most 1.01 and bulk/tail effective sample size at least 400 for every parameter and contrast, with no divergences or saturated tree depths. All eight current posterior fits meet these criteria on the first attempt. Retained draws include chain and iteration identity. Independent density, gradient and contrast reconstruction checks supplement sampler diagnostics.

Table S5 Diagnostics for all eight current posterior fits

Model	Draws	Max R-hat	Min bulk ESS	Min tail ESS	Divergences / depth hits
Common linear original	4000	1.0040	1972	2314	0 / 0
Common linear shifted	4000	1.0068	1245	1618	0 / 0
Common quadratic original	4000	1.0048	1158	1931	0 / 0
Common quadratic shifted	4000	1.0043	1114	2061	0 / 0
Medieval preferred	4000	1.0041	1121	1794	0 / 0
Medieval companion	4000	1.0038	1126	1794	0 / 0
Viking preferred	4000	1.0062	1180	2095	0 / 0
Viking companion	4000	1.0063	941	1428	0 / 0

Current contrasts and sensitivities

Tables S6 and S7 show every Component 3 within-period contrast and every later-profile contrast, including comparisons not discussed in the main text. Online Resource 4 provides all three components and every direct period difference under the four national specifications. Table S8 reports the direct I1–Z284 comparisons on the common Component 3. Conditional intervals exclude uncertainty about the earlier screen, source selection and unknown recruitment probabilities. Different archaeological axes do not acquire a shared historical scale because their fitted coefficients are all standardized.

Table S6 Every current within-period Component 3 contrast

Period and contrast	Linear original	Linear shifted	Quadratic original	Quadratic shifted
Late Neolithic I1–Z284	0.470 [0.087, 0.864]	0.532 [0.190, 0.877]	0.481 [0.110, 0.862]	0.529 [0.172, 0.894]
Late Neolithic I1–U106	−0.098 [−0.427, 0.242]	−0.010 [−0.334, 0.292]	−0.081 [−0.423, 0.248]	0.040 [−0.289, 0.362]
Late Neolithic L22–Z58	0.025 [−0.323, 0.385]	0.239 [−0.108, 0.591]	0.056 [−0.307, 0.410]	0.264 [−0.097, 0.636]
Late Neolithic U106–P312	−0.045 [−0.483, 0.386]	−0.294 [−0.697, 0.114]	−0.068 [−0.526, 0.368]	−0.345 [−0.776, 0.065]
Late Neolithic Z284–U106	−0.569 [−1.020, −0.106]	−0.543 [−0.954, −0.134]	−0.562 [−1.022, −0.110]	−0.489 [−0.917, −0.055]
Late Neolithic N–I1	−0.080 [−0.498, 0.354]	−0.127 [−0.487, 0.239]	−0.101 [−0.508, 0.329]	−0.107 [−0.488, 0.272]
Late Neolithic I1–P312	−0.143 [−0.500, 0.214]	−0.305 [−0.667, 0.027]	−0.149 [−0.496, 0.210]	−0.305 [−0.661, 0.041]
Bronze Age I1–Z284	0.153 [−0.179, 0.486]	0.090 [−0.198, 0.380]	0.150 [−0.182, 0.477]	0.092 [−0.202, 0.394]
Bronze Age I1–U106	0.089 [−0.207, 0.384]	0.091 [−0.189, 0.353]	0.091 [−0.212, 0.395]	0.059 [−0.228, 0.338]
Bronze Age L22–Z58	0.050 [−0.278, 0.376]	−0.194 [−0.499, 0.111]	0.029 [−0.305, 0.368]	−0.210 [−0.516, 0.100]
Bronze Age U106–P312	0.242 [−0.149, 0.619]	0.126 [−0.218, 0.473]	0.237 [−0.140, 0.616]	0.155 [−0.208, 0.512]
Bronze Age Z284–U106	−0.064 [−0.451, 0.329]	0.001 [−0.349, 0.358]	−0.059 [−0.459, 0.351]	−0.032 [−0.408, 0.334]
Bronze Age N–I1	−0.145 [−0.501, 0.218]	−0.051 [−0.383, 0.290]	−0.152 [−0.517, 0.215]	−0.058 [−0.395, 0.294]
Bronze Age I1–P312	0.331 [0.029, 0.635]	0.217 [−0.080, 0.518]	0.329 [0.012, 0.645]	0.214 [−0.076, 0.515]
Roman I1–Z284	−0.257 [−0.679, 0.185]	−0.179 [−0.619, 0.254]	−0.263 [−0.667, 0.166]	−0.173 [−0.621, 0.267]
Roman I1–U106	−0.177 [−0.574, 0.222]	−0.091 [−0.501, 0.321]	−0.187 [−0.591, 0.232]	−0.087 [−0.512, 0.340]
Roman L22–Z58	0.160 [−0.297, 0.615]	−0.078 [−0.566, 0.399]	0.132 [−0.336, 0.590]	−0.091 [−0.567, 0.402]
Roman U106–P312	0.327 [−0.194, 0.850]	0.314 [−0.238, 0.858]	0.363 [−0.195, 0.908]	0.306 [−0.226, 0.843]
Roman Z284–U106	0.080 [−0.413, 0.566]	0.089 [−0.444, 0.622]	0.075 [−0.437, 0.595]	0.086 [−0.438, 0.615]
Roman N–I1	−0.142 [−0.652, 0.376]	−0.401 [−0.895, 0.121]	−0.136 [−0.661, 0.384]	−0.386 [−0.921, 0.137]

Period and contrast	Linear original	Linear shifted	Quadratic original	Quadratic shifted
Roman I1–P312	0.150 [−0.287, 0.593]	0.223 [−0.240, 0.690]	0.176 [−0.288, 0.627]	0.219 [−0.240, 0.669]
Migration I1–Z284	−0.077 [−0.437, 0.276]	−0.066 [−0.451, 0.319]	−0.075 [−0.429, 0.293]	−0.086 [−0.470, 0.329]
Migration I1–U106	0.236 [−0.117, 0.585]	0.039 [−0.306, 0.404]	0.230 [−0.120, 0.587]	0.039 [−0.339, 0.399]
Migration L22–Z58	−0.547 [−0.982, −0.143]	−0.364 [−0.814, 0.065]	−0.536 [−0.958, −0.116]	−0.378 [−0.839, 0.061]
Migration U106–P312	−0.197 [−0.629, 0.236]	0.151 [−0.299, 0.608]	−0.189 [−0.628, 0.261]	0.164 [−0.290, 0.613]
Migration Z284–U106	0.313 [−0.113, 0.748]	0.105 [−0.353, 0.554]	0.304 [−0.131, 0.757]	0.125 [−0.346, 0.589]
Migration N–I1	0.046 [−0.386, 0.468]	0.056 [−0.370, 0.473]	0.058 [−0.369, 0.502]	0.057 [−0.368, 0.481]
Migration I1–P312	0.039 [−0.325, 0.401]	0.191 [−0.190, 0.588]	0.041 [−0.336, 0.418]	0.203 [−0.194, 0.592]

Table S7 Every current lineage contrast on each later profile

Contrast	Viking preferred	Viking companion	Medieval preferred	Medieval companion
I1–Z284	0.066 [−0.193, 0.330]	0.052 [−0.377, 0.485]	0.243 [−0.023, 0.514]	0.067 [−0.195, 0.326]
I1–U106	0.237 [−0.043, 0.524]	0.271 [−0.146, 0.704]	−0.258 [−0.492, −0.020]	−0.427 [−0.706, −0.152]
L22–Z58	0.373 [0.060, 0.681]	0.917 [0.430, 1.397]	−0.057 [−0.327, 0.219]	−0.017 [−0.311, 0.269]
U106–P312	−0.305 [−0.666, 0.056]	−0.525 [−1.062, 0.011]	0.332 [0.030, 0.633]	0.398 [0.055, 0.744]
Z284–U106	0.170 [−0.170, 0.499]	0.218 [−0.311, 0.750]	−0.501 [−0.816, −0.196]	−0.494 [−0.842, −0.155]
N–I1	−0.047 [−0.343, 0.243]	−0.034 [−0.521, 0.451]	0.013 [−0.274, 0.290]	0.125 [−0.147, 0.408]
I1–P312	−0.069 [−0.362, 0.231]	−0.254 [−0.688, 0.187]	0.074 [−0.192, 0.341]	−0.030 [−0.315, 0.250]

Table S8 All direct I1–Z284 period differences on Component 3

Period and contrast	Linear original	Linear shifted	Quadratic original	Quadratic shifted
Late Neolithic minus Bronze Age	0.317 [−0.240, 0.885]	0.442 [−0.036, 0.930]	0.331 [−0.204, 0.862]	0.437 [−0.083, 0.958]
Late Neolithic minus Roman	0.728 [0.100, 1.339]	0.712 [0.119, 1.322]	0.744 [0.138, 1.351]	0.702 [0.097, 1.316]

Period and contrast	Linear original	Linear shifted	Quadratic original	Quadratic shifted
Late Neolithic minus Migration	0.548 [0.007, 1.093]	0.598 [0.013, 1.187]	0.556 [0.021, 1.103]	0.615 [0.003, 1.230]
Bronze Age minus Roman	0.411 [−0.199, 1.007]	0.270 [−0.308, 0.846]	0.412 [−0.197, 1.003]	0.265 [−0.316, 0.858]
Bronze Age minus Migration	0.231 [−0.275, 0.746]	0.156 [−0.329, 0.642]	0.225 [−0.302, 0.727]	0.178 [−0.307, 0.651]
Roman minus Migration	−0.180 [−0.816, 0.489]	−0.113 [−0.790, 0.542]	−0.188 [−0.819, 0.445]	−0.087 [−0.788, 0.577]

The common-period sensitivity analysis comprises 976 conditional fits: 128 paired allocations, 798 crossed-weight replicates, two phase omissions, ten genetic-block omissions, six cell omissions, twelve regional restrictions, ten site omissions, six neighbouring-pair omissions and four paternal-position scenarios. The later-profile sensitivity analysis comprises 1,652 conditional fits: 1,596 crossed-weight fits, 32 block omissions, sixteen cell omissions and eight position scenarios. All planned fits are retained. Successful gradient and curvature checks are numerical diagnostics rather than evidence that the sensitivity scheme has calibrated coverage.

Crossed weights are products of independent Exp(1) block and unit-centroid cell factors (Owen and Eckles 2012), normalized to preserve the total unit weight. The genetic-factor streams are shared where specified; scales, centres and penalties remain fixed. Quantiles across 399 replicates describe dependence sensitivity. They are neither posterior credible intervals nor a generative pedigree model. Whole-group omissions remove all localities of each omitted unit. Archaeological omissions retain paternal units while rebuilding the affected feature and recording measurements.

Single-site nominations use nearest-minus-second-nearest proximity loss. Neighbouring pairs are ranked by the actual joint deletion loss, including replacement by the third-nearest site where the two nominated sites are the nearest alternatives. The candidate domain is the existing nearest-neighbour set within 10 km; three disjoint pairs are retained before inspecting their omission effects. A single-site omission can be small when another nearby site substitutes; disjoint neighbouring pairs test that threat explicitly. Assumed 25 and 50 km origin disks propagate through archaeological measurements and observation controls while nominal geography, terrain and registry controls remain fixed. The disks are scenarios, not measured error bounds.

The current B-26258 omission on Late Neolithic Component 3 gives 0.304 [−0.079, 0.686] and 0.374 [0.019, 0.729] under the two grids, using conditional modes and inverse-Hessian intervals. This block contains twenty Z284 units and twenty-six unit–locality rows and is one of three tied largest blocks. Its twenty-five distinct nominal positions span 58.84–66.10°N and 11.92–22.93°E; their median latitude is 63.18°N. The earlier two-period and current four-period models condition on different archaeological designs, so their omission coefficients answer different comparisons. The primary common-period crossed-weight ranges include zero, whereas the two preferred medieval U106 comparisons retain their directions. The preferred Viking L22–Z58 range includes zero despite positive selected omission modes. These distinctions remain part of the historical assessment.

Ancient evidence and historical triangulation

The reconciled evidence uses sixteen retained original files, including published workbooks (Margaryan et al. 2020; Rodríguez-Varela et al. 2023; Allentoft et al. 2024a, b; Speidel et al. 2025; McColl et al. 2026), Y-chromosome placement summaries and versioned branch dictionaries. All 1,001 retained assertions match their 837 unique original source rows. Repeated use of a source row across questions is not an additional individual. Exact identifiers, original-row identity and explicit published same-person aliases

form 275 identity groups, including 44 documented alias links. Published relatives are not merged. These counts describe the bounded evidence inventory, not an exhaustive ancient census or independent paternal lineages.

The branch crosswalk checks 247 call records representing 202 identities. It retains 155 reported parents or qualified mapped paths. Eighty-two Swedish male calls from the Viking source (Margaryan et al. 2020) are translated through that source's Yleaf 2.1 nomenclature (Ralf et al. 2018, undated) and the fixed YFull version 12.00.0 tree (YFull 2024); 79 yield a qualified path. Literal SNP/path evidence supports the mapping. Labels absent from the dictionary, tied placements and unstable downstream assignments remain qualified. An I1 assignment can remain valid when the terminal branch is unresolved, but R1b-L51/L11/M269 cannot be assigned to U106/P312 without relevant markers and upstream R1a cannot become Z284. Missing labels remain missing, rather than becoming haplogroup N.

The three placement files aggregate.yplace, scoreties_summary.yplace and unstabledownstream.yplace are from the versioned Zenodo dataset of Pinotti and McColl (2025). Published workbooks retain their original article and supplementary-table identifiers in the source catalogue. The Battle Axe supplementary text is also available as the versioned dataset associated with Malmström et al. (2019), https://doi.org/10.6084/m9.figshare.9918416.v2.

Twenty-one original prehistoric context observations and nine contemporary Migration candidates retain normalized date variants with their source columns. Original calibrated bounds and published reservoir-corrected bounds are retained as alternative dating interpretations; neither is replaced by an assumed uniform date distribution. Context-dated observations are labelled accordingly. NEO220 retains original 2137–1940 BCE and reservoir-corrected 2140–1932 BCE bounds; NEO223 retains original 2134–1916 BCE and corrected 2135–1900 BCE bounds with its low-coverage/low-genotype-probability flags. Ancient dates in this inventory are published source values, not recalibrations through the archaeological catalogue model.

The source-selected McColl et al. (2026) geographical layer contains 135 Swedish clustered observations. Seventeen fall nominally in the 2300–500 BCE interval, and sixteen belong to the two nominated ancestry clusters. In the Late Neolithic selection, the eastern cluster has eight observations at three coordinates and the southern cluster seven at four, with non-overlapping latitude ranges. The Bronze Age selection has one eastern observation and no southern comparator. These autosomal clusters do not substitute for paternal branches. A model of their landscape scores would largely restate their geographical separation, so no adjusted ancient association is fitted.

The contemporary Migration collection (Rodríguez-Varela et al. 2023) has nine context-dated candidates, seven retained in the source study. Its three retained broad-I1 observations occupy two coordinates; relatedness and two source exclusions remain attached. Under the specified branch rules there is no qualified contemporary Z284/U106 comparison group. The separately qualified Z284 endpoint inventory contains five identities at three coordinates, but does not provide the missing contemporaneous comparison group. These are limitations of the retained comparison, not proof that those branches were absent from Sweden.

The inventory retains every individual and records their eligibility for each period comparison. Plotting uses the reconciled person-level branch outcome: incompatible qualified paths or broad parents remain unresolved. Published low-coverage flags, original-call caveats, population-analysis exclusions and reported kinship have separate fields; missing status is not a quality pass and no universal coverage cutoff is imposed. Öllsjö oll009 retains its original tentative I1 qualification (Malmström et al. 2019), separately from later branch labels. The strict direct-dating screen assesses whether a comparison is feasible; it does not establish independence among individuals. CGG015491, CGG105935 and CGG106210 remain explicit metadata-only cases. CGG105930 retains the source mean AD 135 outside its reported AD 140–200 bounds, with the inconsistency flagged; overlap uses the reported bounds. CGG105925, whose contextual fields give [AD 100, AD 100], is included in the Roman window without treating its event date as exact.

Source inclusion fields are normalized as included, excluded or unknown using the stated source conventions. FALSE followed by a reason is retained as exclusion in the Cell 2023 population-analysis field; missing values remain unknown. Published contMT_5pct and first-degree relationship tokens retain their raw text and row locators. Quality, call ambiguity, kinship and merged-sample identity are separate qualifications. In particular, lov002 and rtp004 carry source-exclusion markers, NEO226 carries its mitochondrial-contamination caveat, and VK517 retains its reported relationship to VK527. These annotations do not change modern-lineage model inputs.

The northern alternative combines the DATASET05.xlsx compilation of Hennius (2022), associated with Hennius (2020), and Tables 3–4 of Rimpi (2025). It retains 439 assertions, 353 Swedish and 86 Norwegian. Of the Swedish assertions, 255 link to 105 KMR objects. Those objects do not establish 105 independent, directly dated trapping systems. The available phase and comparator evidence does not support the proposed historical link to the fitted southern and central sequence. Broad Iron Age N–I1 points toward occupation-related evidence rather than trapping; the Neolithic trapping contrast concerns L22–Z58. Neither the N branch label nor a taiga setting supplies an ethnic attribution. The candidate remains documented without a new model unsupported by its chronology.

Paired chronology and source-pool sensitivity

The earlier posthole–hearth comparison uses four discontinuous windows: 4000–2800 BCE, 2300–1700 BCE, 1700–1100 BCE and AD 400–550. These differ from the four windows of the current seven-family basis. Complete containment of a material's 95.4% envelope originally defined the focal panels. Primary posthole/hearth site counts are 61/75, 62/38, 45/92 and 29/68. Annual-material and expanded lower-inbuilt-age restrictions do not support every window; in particular, Late Neolithic hearth support remains inadequate. A failure to fit the restricted panel is not a negative effect estimate.

The paired appraisal fixes the original 1,309 recording assertions and their focal roles, then compares them with an expanded 12,179-assertion pool. It retains source exclusions, corrected measurements and site identities; the restored Fullerö assertion contributes recording information while remaining excluded as a generic hearth. Each of 64 laboratory-date allocations is shared across the two pools. Within an allocation, the measurement averages the same 64 archaeological-position realizations. This differs from the common-family procedure, which pairs one date draw with one position draw. Both preserve the paternal mixture and frozen transformations.

There are 256 allocation fits, comprising two pools, two grids and 64 allocations, plus two original-panel mode checks. All sixty original lineage and temporal contrasts per allocation are retained. Variation across the conditional modes describes measurement sensitivity rather than posterior uncertainty. The allocation procedure does not estimate material-to-event offsets. The principal empirical contrast is that within-pool date allocation largely preserves the Migration association, whereas expanding the pool changes its magnitude substantially.

Table S9 Paired I1–Z284 date-allocation and source-pool sensitivity. Medians summarize 64 date allocations; the final two columns count allocations whose inverse-Hessian 95% interval lies entirely above zero.

Period / grid	Original median	Expanded median	Original: lower bound > 0	Expanded: lower bound > 0
4000–2800 BCE / Original	0.050	0.006	0/64	0/64
4000–2800 BCE / Shifted	0.024	0.009	0/64	0/64
2300–1700 BCE / Original	0.242	0.249	1/64	32/64

Period / grid	Original median	Expanded median	Original: lower bound > 0	Expanded: lower bound > 0
2300–1700 BCE / Shifted	0.249	0.262	2/64	38/64
1700–1100 BCE / Original	−0.196	−0.090	0/64	0/64
1700–1100 BCE / Shifted	−0.134	−0.093	0/64	0/64
AD 400–550 / Original	0.349	0.047	62/64	0/64
AD 400–550 / Shifted	0.412	0.062	64/64	2/64

The original-grid Kvarnbacken single-site omissions (Helander 2013) illustrate spatial substitution: removing either site separately gives a mode of about 0.352, compared with baseline 0.350, while removing both gives approximately 0.244. Full posterior follow-ups for both grids retain their original source panels and are separate from the current common-family omissions. The annual cereal samples demonstrate good material identification without uniquely dating building construction. The selected material and building context therefore remain distinct evidential statements.

The full earlier appraisal retains ten posterior fits: baseline, covered-cohort, quadratic-geography, joint Kvarnbacken omission and Bobygden omission (Altner 2014), each under both grid origins. Coverage requires every origin of a retained unit to have both Migration feature proximities at least exp(−2), a proximity-equivalent 100 km rule. The material-quality variant remains unfitted because its prespecified archaeological support criteria fail. These checks retain the earlier four-window measurement and do not replace the common seven-family analysis.

Estimator performance and chronological discrimination

The first benchmark and chronological benchmark retain their original generating designs and conditions. Their methods are reported below because the numerical claims concern these specific evaluations. They do not validate a new source correction by analogy, repeat the axis search, or establish the calibration of the current common-family and later-profile posteriors. All adverse conditions and predictive comparisons remain included.

Spatial structure and sampling can affect population-genetic inference (Battey et al. 2020). The present evaluation retains the observed Swedish design and follows an explicit specification of generating conditions, estimand, methods and performance measures (Morris et al. 2019). All empirical findings were known before this specification. The injected contrast of 0.4 was rounded from preliminary Swedish posterior estimates of approximately 0.37 and 0.43; its choice was informed by the empirical results.

The original regional grid, 1,335 units, 1,594 locality rows, 246 block memberships, four archaeological exposures and all covariates remain fixed. A constrained fit sets all archaeological slopes to zero while estimating the remaining coefficients under the unchanged likelihood and priors. These nuisance estimates are used once, without drawing them from their posterior. Unit probabilities average the locality-specific eight-category probabilities with the same equal locality weights used in estimation. The signal condition adds a Migration slope of +0.2 to each I1 category and −0.2 to Z284, leaving the other categories and period slopes unchanged. Subtracting the reference-category coefficient preserves these contrasts and gives a weighted I1–Z284 difference of exactly 0.4.

The retained simulation design and all 2,000 completed fits predate the subsequent 358 m correction to the Örtofta site location. The complete generating arrays and parameters are retained in data/benchmarks/, and

the benchmark command reuses them. Final empirical fits use the corrected archaeological layer. No simulation condition or seed was changed in response to the correction or later appraisal.

Independent draws and correlated draws are each generated with and without the injected contrast. For correlated draws, a Gaussian copula supplies one latent standard-normal draw per retained block and an independent draw per unit, each multiplied by the square root of 0.5. Transforming their sum by the standard-normal distribution function and then the unit-specific categorical cumulative distribution preserves the marginal unit probabilities. Category order is fixed to that in Table S1. The resulting dependence is a stress condition imposed on existing group geometry; it is not a fitted pedigree process. Blocks were defined from the empirical data, and their reuse does not constitute independent validation of those definitions.

A fifth condition has no direct archaeological effect and adds a fixed category-specific spatial field generated from standardized squared northing, northing times easting, and squared easting. Coefficients are drawn once from $N(0, 0.5^2)$, using seed 2026110608, and reference-category values are subtracted. No archaeological measurement enters selection of this field. The five conditions use 200 datasets each and simulation seed 2026110607. The observation footprint is fixed; archaeological recording and excavation are not independently generated. Under omitted geography, comparison to the injected zero effect measures apparent archaeological association in its absence; it does not assert that the misspecified fitted model has a zero population coefficient.

Each dataset is fitted with the full regional model and a simpler model retaining the same cell-centred exposures, likelihood, priors, intercept, linear coordinates, recruitment and terrain. The simpler model omits regional intercepts and all eight period-specific recording and opportunity controls. This paired comparison assesses removal of those adjustments. Because the generating model retains them, it is not an unrestricted comparison with competing spatial methods. Both analyses use conditional modes and inverse-Hessian intervals; neither repeats posterior sampling or crossed-weight reweighting.

All 2,000 fits have a maximum absolute gradient below 0.00001 and positive local curvature. Table S10 reports bias, root mean square error, interval coverage and zero exclusion. Rate intervals are 95% Wilson intervals; Monte Carlo standard errors for bias, empirical standard deviations, mean reported standard deviations and paired squared-error differences are retained in the database. No failed fit or unfavourable condition was discarded. The full model has smaller absolute bias in every condition, while the simpler model has slightly lower root mean square error in the independent signal condition. The benchmark does not estimate the false-positive probability of the selected empirical finding or calibrate the earlier axis search.

Table S10 All conditions in the first fixed-design benchmark

Condition	Model	Bias	RMSE	Coverage (%)	Excludes zero (%)
Independent, β = 0	Full	−0.001	0.139	98	2
Independent, β = 0	Simple	−0.046	0.147	93.5	6.5
Independent, β = 0.4	Full	−0.004	0.131	92.5	86
Independent, β = 0.4	Simple	−0.045	0.126	94.5	82.5
Correlated, β = 0	Full	−0.005	0.129	95.5	4.5
Correlated, β = 0	Simple	−0.053	0.140	94	6
Correlated, β = 0.4	Full	+0.011	0.119	97	90.5
Correlated, β = 0.4	Simple	−0.035	0.124	95.5	85.5

Condition	Model	Bias	RMSE	Coverage (%)	Excludes zero (%)
Curvature, β = 0	Full	+0.008	0.125	98	2
Curvature, β = 0	Simple	−0.050	0.133	95	5

The generating nuisance coefficients are estimated once under the earlier posthole–hearth original-grid model with all four archaeological slopes fixed to zero. The same 1,335 units, equal-origin mixture likelihood, eight categories, priors and reference transformations are used throughout. The category direction is +0.5 for each I1 category, −0.5 for Z284 and zero otherwise, with the reference-category value subtracted. Thus every I1–Z284 slope difference equals the specified injection. Scenarios comprise a global null (1,000 datasets), a 0.2 or 0.4 contrast in each period separately (eight conditions, 400 each), equal 0.2 contrasts across all periods (400), a pooled-landscape 0.4 contrast (400), and selective-recording null and Migration 0.4 conditions (400 each). Training and independent test category draws use seeds 2026110617 and 2026110618, indexed by condition and realization. They are generated at the same observed profiles; this is not a held-out-region experiment.

The three fitted models share all nuisance controls and their priors. They differ only in having no focal archaeological term, a pooled posthole–hearth term, or the four period terms. Pooled proximity takes the nearest feature in the union of eligible period panels within each positional realization, then averages; it is not the sum of period exposures. Its global and regional centres are frozen on the reference cohort and it uses the same common exposure scale. The complete four-slope covariance generates simultaneous intervals for four slopes and six pairwise differences, using a maximum-standardized-contrast construction (Hothorn et al. 2008). The critical value is the 95th percentile of the maximum absolute standardized contrast from 16,384 fixed scrambled Sobol draws of four standard normal variables, transformed by that covariance; seed 2026110619 fixes integration. A unique positive-period nomination requires a positive slope lower bound and positive lower bounds for all three differences, plus at least five distinct sites for each feature in every compared period. Otherwise decisions distinguish positive but unresolved chronology, no detected positive relationship, and insufficient archaeological support. Unresolved differences do not establish equivalence. This rule was evaluated prospectively within this module and was not retroactively applied to the empirical Bayesian intervals.

The recording stress defines a hypothetical latent panel by the period with the largest calibrated material-date probability, provided the outer 95.4% envelope intersects a studied window. This is simulation truth, not a recovered event chronology. Connected source groups join canonical evidence groups when a laboratory determination is duplicated. One shared uniform draw per group determines retention with period- and feature-specific base probabilities: posthole/hearth pairs are 81/148 and 91/134; 101/247 and 39/120; 63/262 and 103/469; and 41/1098 and 72/1481. Other contexts use the mean of the pair. The logit additionally includes 0.7 times standardized group northing. Group northing is the mean over distinct sites, standardized without weights across groups. Outcomes arise from latent-panel measurements; fitted measurements and observation controls are rebuilt from the retained evidence with fixed scales. Empty panels yield zero proximity and fail support. Actual retained counts accompany every fit. The process tests one specified form of selective recording; it does not simulate the full radiocarbon, excavation or publication process.

Performance separates coefficient recovery, temporal discrimination and prediction. The four slopes retain bias, root mean square error and marginal interval coverage; all ten contrasts retain joint coverage. Decision rates use all scheduled datasets as denominators and include failures and support failures. Expected and independently sampled log and multicategory Brier scores are paired across methods (Gneiting and Raftery 2007); probability-bin calibration compares fitted and generating probabilities. Wilson intervals quantify rate uncertainty and replicate-level standard errors quantify Monte Carlo uncertainty in means and paired differences. Under the pooled generator there is no declared true vector of

four period slopes: coefficient bias and correct-period claims are therefore omitted. Under recording stress, recovery refers to the injected latent contrast, not a guaranteed population coefficient in the misspecified observed-panel model. Full individual results, all four estimates and covariance matrices, calibration bins and all adverse outcomes remain in the database.

Table S11 Chronological decisions and joint coverage in every benchmark condition

Condition	Any nonzero slope (%)	Correct unique period (%)	Wrong unique period (%)	Joint coverage (%)	Insufficient support (%)
No effect	2.3 [1.5, 3.4]	0.0 [0.0, 0.4]	0.0 [0.0, 0.4]	95.4 [93.9, 96.5]	0.0 [0.0, 0.4]
4000–2800 BCE, 0.2	22.2 [18.4, 26.6]	0.5 [0.1, 1.8]	0.0 [0.0, 1.0]	96.2 [93.9, 97.7]	0.0 [0.0, 1.0]
Late Neolithic, 0.2	13.8 [10.7, 17.5]	0.5 [0.1, 1.8]	0.0 [0.0, 1.0]	94.2 [91.5, 96.1]	0.0 [0.0, 1.0]
Early Bronze, 0.2	9.0 [6.6, 12.2]	0.2 [0.0, 1.4]	0.0 [0.0, 1.0]	94.2 [91.5, 96.1]	0.0 [0.0, 1.0]
Migration, 0.2	16.0 [12.7, 19.9]	0.8 [0.3, 2.2]	0.0 [0.0, 1.0]	95.2 [92.7, 96.9]	0.0 [0.0, 1.0]
4000–2800 BCE, 0.4	89.2 [85.8, 91.9]	14.8 [11.6, 18.6]	0.0 [0.0, 1.0]	96.8 [94.5, 98.1]	0.0 [0.0, 1.0]
Late Neolithic, 0.4	65.2 [60.5, 69.8]	13.2 [10.3, 16.9]	0.0 [0.0, 1.0]	96.5 [94.2, 97.9]	0.0 [0.0, 1.0]
Early Bronze, 0.4	52.5 [47.6, 57.3]	5.8 [3.9, 8.5]	0.0 [0.0, 1.0]	93.8 [90.9, 95.7]	0.0 [0.0, 1.0]
Migration, 0.4	66.5 [61.7, 70.9]	10.0 [7.4, 13.3]	0.0 [0.0, 1.0]	96.0 [93.6, 97.5]	0.0 [0.0, 1.0]
All periods, 0.2	43.8 [39.0, 48.6]	0.0 [0.0, 1.0]	0.0 [0.0, 1.0]	98.2 [96.4, 99.1]	0.0 [0.0, 1.0]
Pooled landscape, 0.4	38.2 [33.6, 43.1]	Not defined	Not defined	Not defined	0.0 [0.0, 1.0]
Recording stress, null	3.5 [2.1, 5.8]	0.0 [0.0, 1.0]	0.0 [0.0, 1.0]	95.8 [93.3, 97.3]	0.0 [0.0, 1.0]
Recording stress, Migration 0.4	3.5 [2.1, 5.8]	0.0 [0.0, 1.0]	0.0 [0.0, 1.0]	40.8 [36.0, 45.6]	0.2 [0.0, 1.4]

Rates use every scheduled dataset in the condition; bracketed limits are 95% Wilson intervals. A correct unique-period decision requires a positive simultaneous slope interval and positive simultaneous differences from all three other periods, with adequate archaeological support. “Not defined” denotes a structurally undefined decision under the generator, rather than an observed zero rate. Joint coverage concerns all four slopes and six direct differences.

Table S12 Paired predictive performance under matched observation controls

Condition	Log gain vs geography	Log gain vs pooled	Brier gain vs geography	Brier gain vs pooled
No effect	−10.46 (0.09)	−7.75 (0.09)	−2.77 (0.03)	−2.02 (0.02)
4000–2800 BCE, 0.2	−8.53 (0.15)	−6.63 (0.15)	−2.24 (0.04)	−1.74 (0.04)
Late Neolithic, 0.2	−8.89 (0.14)	−6.43 (0.14)	−2.34 (0.04)	−1.65 (0.04)
Early Bronze, 0.2	−9.29 (0.14)	−6.87 (0.14)	−2.49 (0.04)	−1.80 (0.04)
Migration, 0.2	−9.60 (0.14)	−7.03 (0.14)	−2.54 (0.04)	−1.84 (0.04)
4000–2800 BCE, 0.4	−3.59 (0.16)	−3.68 (0.15)	−0.83 (0.04)	−0.94 (0.04)
Late Neolithic, 0.4	−4.66 (0.14)	−2.95 (0.14)	−1.25 (0.04)	−0.75 (0.04)

Condition	Log gain vs geography	Log gain vs pooled	Brier gain vs geography	Brier gain vs pooled
Early Bronze, 0.4	−5.69 (0.15)	−3.72 (0.15)	−1.55 (0.04)	−0.97 (0.04)
Migration, 0.4	−6.26 (0.15)	−3.74 (0.15)	−1.66 (0.04)	−0.97 (0.04)
All periods, 0.2	−3.05 (0.14)	−4.23 (0.14)	−0.82 (0.04)	−1.12 (0.04)
Pooled landscape, 0.4	−6.57 (0.14)	−10.33 (0.14)	−1.55 (0.04)	−2.71 (0.04)
Recording stress, null	−9.66 (0.13)	−7.16 (0.13)	−2.57 (0.04)	−1.87 (0.04)
Recording stress, Migration 0.4	−9.42 (0.15)	−7.04 (0.14)	−2.54 (0.04)	−1.86 (0.04)

Values and parenthesized Monte Carlo standard errors are multiplied by 1,000. Log-score gain is four-period model minus comparator; Brier-score gain is comparator minus four-period model. Positive values favour the four-period model. Scores are averaged over units; log scores use natural logarithms and the multicategory Brier score sums squared probability errors over all eight categories without dividing by eight. Expected scores integrate the generating category probabilities; sampled scores use independently simulated labels. Paired differences use the same simulated datasets.

Sensitivity to phylogenetic resolution

The branch-cut algorithm counts the initially deepest observed screened labels below each node of the frozen tree. It then moves each screened record's label to the highest ancestor in the same outcome category with at most K such descendant labels, with K fixed at 2 or 5. A label is retained if no eligible ancestor exists. The same procedures are then used for non-nested pruning, documented-family merging, unit construction, whole-unit terrain exclusion and proxy dependence grouping. Existing regional exposure centres remain fixed; any newly represented cell receives a centre calculated before effects from equally weighted screened source-record localities. The sum-to-zero regional basis expands with the same prior. Composite I1 weights remain 237/539, 207/539 and 95/539. The four posterior jobs use the standard four-chain diagnostic and single-retry rules, with seeds 2026110621 + K + grid origin. Every coefficient and all 60 contrasts per fit are retained. This sensitivity examines alternative branch representations; it does not establish independent pedigrees or recover unobserved descendants.

Table S13 Earlier posthole–hearth sensitivity to upward phylogenetic cuts

Representation / grid	Units	Records	Blocks	Migration I1–Z284	Max R-hat
K = 2 / Original	1014	2130	202	0.383 [0.061, 0.720]	1.0084
K = 2 / Shifted	1014	2130	202	0.458 [0.144, 0.781]	1.0074
K = 5 / Original	633	2393	133	0.484 [−0.025, 1.005]	1.0066
K = 5 / Shifted	633	2393	133	0.519 [0.066, 1.001]	1.0070

The two-descendant cut retains 1,014 units and 2,130 records, including 495 records absent from the primary cohort. The five-descendant cut retains 633 units and 2,393 records, with 768 recovered records. Six and sixteen primary records respectively cease to be represented. The increase in retained records accompanies stronger aggregation and wider uncertainty. These earlier posthole–hearth results concern alternative represented branches, not independent genealogical families and not current common-profile refits.

Source interpretation and coordinate inventory

Table S14 reports every site in the fixed clearance review: fifteen qualifying in the original reference measurement and three additional positive nearest-gap contributors, marked †. These are original review identifiers, preserved through subsequent site corrections; they are not a count of qualifying sites in the current primary panel. Source fields, assertion identifiers, laboratory links and page locators are retained in data/reports/reviews.json. Page numbers refer to original PDFs unless labelled as printed pages. The accompanying sources.json maps excerpt pages to original pages and supplies report URLs and file hashes; differing printed page numbers remain in the reviews. The four contexts with insufficient source information remain unresolved.

Table S14 All eighteen nominated Late Neolithic clearance contexts

Reference site	Report and PDF pages	Activity interpretation and temporal limit
L1953:8049	Öjabymotet (Emilsson et al. 2020) PDF 70, 71, 80, 81, 168	Hazel charcoal derives from a probable older ground horizon beneath A321. Pollen indicates an open woodland/pasture mosaic, without cereal pollen in this context; it does not date Late Neolithic cultivation. The documented source corrections address discrepancies in catalogue context, species and selected assay links.
L1954:1259	Räppe (Lindman 2001) PDF 14, 15, 16, 17	Cultivation is supported in the wider area, but not at the nominated Late Neolithic date. The report gives Ua-25792 as 2635 ± 90 BP; the catalogue gives Ua-25792B as 3635 ± 90 BP. The unexplained suffix and 1000-year discrepancy remain unresolved; both reported ages are retained.
L1954:1841 †	Rödje (Färjare 2021) PDF 11, 12, 13, 14, 16, 24, 27	Hazel charcoal beneath A133 is 3343 ± 32 BP in the laboratory certificate. Clearance and cultivation soil support land use, principally beginning in the Early Bronze Age; the charcoal may predate clearance. A small Late Neolithic probability tail is not an independently identified field phase.
L1954:3227	Hamneda (Torstensdotter Åhlin et al. 1998) PDF 38, 41, 42, 43, 44, 48, 65	Ua-6593, 3400 ± 70 BP, is verified from cairn A301. The report interprets its unusually early charcoal as probably earlier settlement activity; most cairn activity is much later. Earlier hearths and graves do not establish cultivation contemporary with this charcoal.
L1957:3081	Kölbygärde (Ring et al. 2001) PDF 42, 61, 62, 63	Late Neolithic oak charcoal occurs beneath OL62, separated from the cairn by later charcoal. Crop remains and pollen support Bronze Age cultivation, but the source does not securely date the relevant pollen phase to the Late Neolithic. Inbuilt age and reworked deposits limit event attribution.
L1964:3336	Kråkhult (Hellgren 2023) PDF 15, 24, 25, 32, 37, 39, 40	Sparse alder charcoal, 3459 ± 31 BP, supports early activity. The report prefers grazing improvement from its context and lack of nearby settlement evidence; grazing is an interpretation, not a direct observation. Conflicting A1402/A1872 context labels remain within one site. No contemporary crop evidence is demonstrated.
L1971:8017	Farstorp (Petersson 2015) PDF 54, 55, 140, 142, 152, 153, 179	Late Neolithic pine charcoal occurs in a long-used soil profile beside a later cairn. The report describes older charcoal incorporated into younger features by cultivation. Micromorphology, pollen and stratigraphy support later land use; they do not date cultivation to the charcoal age.
L1976:5349 †	Original report not verified	The original context could not be verified in this review. The catalogue classification is retained, but the activity and its relationship to the dated material remain unresolved. This is a gap in the source evidence.
L1980:4822	Original report not verified	The original context could not be verified in this review. The catalogue classification is retained, but the activity and its relationship to the dated material remain unresolved. This is a gap in the source evidence.

Reference site	Report and PDF pages	Activity interpretation and temporal limit
L1988:8595	Mölletofta and appendix (Andersson 2004) PDF 5, 6, 7, 8, 9, 13	A200 has Ua-8263, 3910 ± 60 BP. Earlier burning or small-scale clearance is possible, preceding the main Roman-to-medieval cairn formation. Soil chemistry supports agricultural use without assigning it to this determination. One date does not date all use of a multi-phase structure.
L1990:5223 †	Original report not verified	The original context could not be verified in this review. The catalogue classification is retained, but the activity and its relationship to the dated material remain unresolved. This is a gap in the source evidence.
L1990:5293	Carlie (2002), printed pp. 305–306, 310–312; Lagerås (2002), printed pp. 403, 405–406, 411	Birch charcoal from A910b and A1401 precedes the main late Bronze/Pre-Roman clearance sequence. Pollen and regional settlement evidence support later agrarian expansion, while early clearance burning remains distinct from crop cultivation and cairn construction. Some soil charcoal can be unrelated to either activity.
L1990:6000	Carlie (2002), printed pp. 312–313, 326–328; Lagerås (2002), printed pp. 403, 405–406, 411	Early charcoal in A704 and A1259 is compatible with clearance burning around the Late Neolithic/Early Bronze transition. The main cultivation and stone-clearance expansion is later. Regional pollen and crop evidence do not independently date each early cairn. This shares the Arkelstorp project and synthesis with the preceding case.
L1996:7017	Mannaberg (Nordin 2023) PDF 19, 20, 36, 67, 97, 100	The report supports a cultivation setting in A1462: a dark homogeneous horizon, charcoal and brushwood, one unidentified cereal grain and nearby Neolithic buildings. Ash charcoal is 3449 ± 30 BP. The grain is undated; older pottery, household ash and repeated use prevent equating material date with cultivation.
L2008:2467	Glasskärvan (Petersson and Zetterlund 2021) PDF 17, 18, 30, 32	Linden charcoal, 3408 ± 34 BP, comes from a pit or lens below a clearance cairn. An overlying cultivation layer has an older date, so a simple construction sequence fails. Later cultivation and settlement are supported; older charcoal is activity evidence with uncertain event linkage.
L2008:419	Bådstorp main and appendices (Seiler and Sjölin 2022) PDF Bådstorp main (2022): 107, 109, 234, 235; Bådstorp appendices (2022): 343	Charcoal dated 3440 ± 33 BP occurs in a historically dated cultivation layer. The report interprets it as probably older natural-fire material. Nearby Bronze Age and historical buildings/crops do not date cultivation to that charcoal. The source-qualified site-identity correction is retained separately from the original identifier.
L2010:9040	Väderstad (Ericsson and Franzén 2005) PDF 49, 55, 58, 59, 75, 83, 84	Early hazel/linden charcoal beneath cairns or clearance layers may reflect burning before the later Bronze Age settlement and cultivation phase. An isolated early oak date in A6001 does not date a predominantly medieval sequence. Cultivation, meadow and pasture remain distinct possible uses.
L2012:8544	Original report not verified	The original context could not be verified in this review. The catalogue classification is retained, but the activity and its relationship to the dated material remain unresolved. This is a gap in the source evidence.

Table S15 documents the evidence for both sides of the later profiles, including cases where a proposed contextual link could not be verified. Charcoal production does not itself establish a local bloomery, and an oven can be a brick kiln. The inconsistent feature–date links in the Selånger report are recorded, and the specified assertions are withheld; unrelated houses in the same report cannot substitute for postholes meeting a strict period-membership rule. Full report-part locators and the source adjudications remain attached.

The report catalogue also retains Lindman's (2003) clearance investigation, the cover identifying Petersson's (2001) report, and source checks against Ericson (2007) and Henriksson (2023). Each retained

excerpt has an original-report page mapping; appendices are attributed to their parent reports rather than counted as independent publications. SOURCES.tsv includes the original URLs for the full 27-excerpt catalogue and four additional reports in the archaeological source archive.

Table S15 Original-report qualification of the later functional profiles

Report and assertion	Verified context	Interpretation and limit
Färila (Altner 2016) C14:21943	Ua-51369: A1 upper charcoal layer, pine, 1140 ± 30 BP; companion Ua-51370 lower layer, 1174 ± 29 BP. Rectangular charcoal-production pit Färila 15:3 in the larger registered area; three stratified burning episodes.	Secure charcoal-production function and broad Viking-period use; not a demonstrated local bloomery. Search around the pit found no iron-production remains. Pine inbuilt age and repeated use; two samples do not equal independent sites. Charcoal had multiple possible consumers.
Harbo-Smedsbo (Kjellberg 2006) C14:18104	Ua-32214: sample 2380, slag heap A2359, Area C, aspen/Salix charcoal, 1135 ± 40 BP, reported 780–990 CE at two sigma.	Iron-production waste is supported by slag/furnace-material analyses; the sampled heap is not itself a furnace. Waste deposition and charcoal age do not date every production episode. Possible stone setting overlies the heap; the site is multi-period.
Fyrislund (Lucas and Lucas 2013) C14:20197	Ua-41825: hearth A6868, pine charcoal, 1174 ± 30 BP, reported 770–970 CE.	A settlement combustion context, distinct from charcoal manufacture. House 15 has domestic remains and two broad late Viking/early medieval dates; House 19 has a proposed dated entrance post. The focal hearth is not thereby assigned to a specific house. House 15 dates and House 19 entry-post dates do not qualify under the inherited strict Viking panel; they may illustrate independent context but cannot validate the included posthole pole by substitution.
Selånger (Strucke and Bäck 2003) C14:14111, C14:14112, C14:14113	Barley dates exist; their feature/context and building-age interpretation are contradicted or unresolved within the original report.	Evidence of earlier cereal material in a later, multi-period setting; not securely dated Viking house construction or use. See six-assay adjudication ledger. Radiocarbon table is on appendix PDF 13 (printed 89); pit plan on middle appendix PDF 15 (printed 73).
Leksand (Holmqvist 2021) C14:571, C14:572	Context 104 charcoal from tar production: 20C/1195 sample 104:44, 610 ± 30 BP; 20C/1196 sample 104:46, 760 ± 30 BP.	Medieval tar-production structure L1998:6741, with construction and channel described. The adjacent charcoal pit belongs to a much later period, so the tar dates cannot be applied to the whole site. The sequence of production episodes remains uncertain.
Nifsarp (Wennerberg 2008) C14:18571, C14:18572	Charcoal-production pit A7, pine: lower sample PK885/Ua35051 and upper PK886/Ua35052. Both medieval; apparent age order reverses stratigraphy.	Production and reuse are better supported than precise episode ordering. Possible inbuilt age; stratigraphy overrides an assumed chronological ordering of the two radiocarbon central estimates.
Uppsala brick kilns (Fagerlund 2016) C14:22096	Ua-51656, PK10173/A817 (S817), 503 ± 33 BP; birch charcoal associated with lower brick course S1458.	Brick-kiln complex with early fifteenth-century construction-related material, not a uniformly domestic oven. Birch inbuilt age, multiple construction phases; no national inference of ecclesiastical production or urban migration from this example.

Online Resource 2, archaeological_coordinates.tsv, contains 9,045 uniquely keyed entries: 4,002 site representatives from the common-period and later pools and original-context reviews, 4,894 associated geometry representatives, and 149 ancient observations reconciled across sources in the mapping dataset. Period filters determine which ancient observations appear in each map. Different entity types are not pooled as independent archaeological sites. The separate confidential reviewer inventory adds 1,404 participant-linked paternal-origin positions, for 10,449 entries. It distinguishes 1,404 measured positions

from 1,399 positions used in the common-period likelihood. Ancient coordinates identify published archaeological locations, not established birthplaces or lifetime residences.

The inventory preserves source decimal coordinates and adds degrees–minutes–seconds formatted to 0.1 arcsecond. This conversion introduces no new claim of accuracy. Each row states the geographical object represented, a source locator and its reliability limits; registry rows retain declared uncertainty where available. Archaeological geometry points are the representatives used in measurement, rather than nearest polygon boundaries or surveyed sample coordinates. Source rounding, geometry extent, recorded positional uncertainty and the material-to-site link remain separate. Figure maps use SWEREF 99 TM; the coordinate inventory expresses latitude and longitude in the retained geographical system.

Regional support and shared recording opportunity

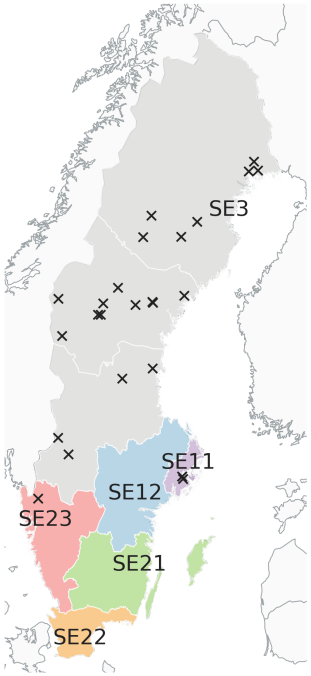
Figure S1 shows estimates for the six regional subsets under both grid origins. SE11 is Stockholm, SE12 eastern central Sweden, SE21 the Småland and Islands region, SE22 southern Sweden, SE23 western Sweden, and SE3 the combined northern groups SE31–SE33. All original estimates are retained. Grey indicates limited support in at least one constituent I1–Z284 category: fewer than ten units or three proxy dependence blocks. This affects presentation, not which estimates were fitted. Table S16 provides complete category-level support. The remaining 71 mixed-region or unassigned units remain in the national analysis. Region definitions: Eurostat (2024), GISCO NUTS 2024; © EuroGeographics for the administrative boundaries.

Table S16 Regional units and proxy dependence blocks in every outcome category

Region	L22	Z58	Other I1	Z284	U106	P312	N	Other
SE11	5/4	5/5	4/4	5/4	5/5	1/1	7/5	6/6
SE12	50/25	28/16	15/10	19/13	27/16	20/10	34/13	39/19
SE21	41/23	25/14	17/9	15/8	25/17	16/8	2/2	30/18
SE22	39/20	19/10	3/3	6/4	20/11	24/13	1/1	47/25
SE23	34/19	12/8	24/8	25/10	30/15	33/10	9/7	43/19
SE3	57/21	106/24	27/10	79/17	42/24	39/14	64/17	40/15

Cells show units/blocks; one block may appear in several categories or regional restrictions. Composite I1 retains the frozen 237:207:95 weights. Total blocks are therefore not sums of category counts.

a **Regional support and the influential paternal block**

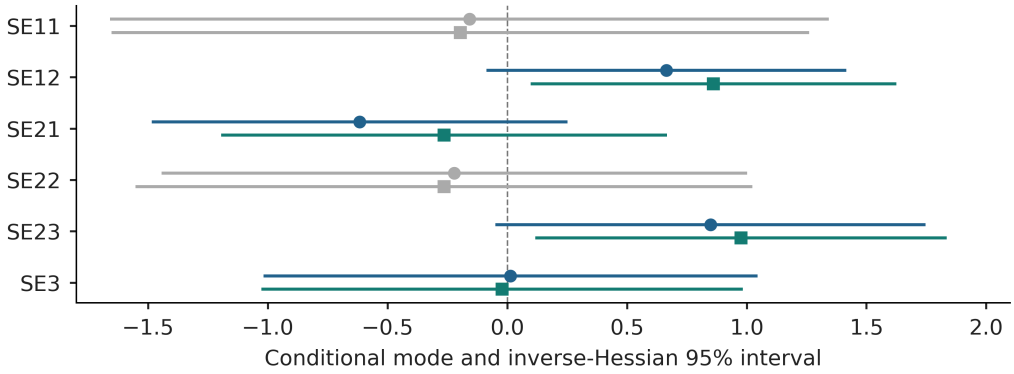


Group	Units	Blocks	I1	Z284
SE11	38	34	14	5
SE12	232	122	93	19
SE21	171	99	83	15
SE22	159	87	61	6
SE23	210	96	70	25
SE3	454	142	190	79

SE3 combines SE31, SE32 and SE33.
71 mixed or unassigned units are outside these regional restrictions; they remain in the national fit.

× B-26258: 20 units; 25 distinct positions
58.84–66.10°N; median latitude 63.18°N

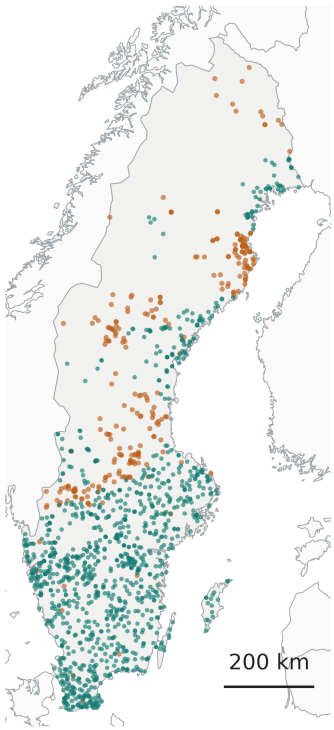
b **Regional Late Neolithic I1–Z284 estimates**



Circles: original grid; squares: shifted grid. Grey: fewer than ten units or three blocks in a constituent I1–Z284 category. These are six restricted cohorts, each fitted twice. Complete category-level unit and block counts accompany the numerical inventory.

Fig. S1 Regional support and the influential B-26258 block. a Frozen regional groups, represented units and proxy dependence blocks; crosses locate the block's 25 distinct positions. b All twelve regional Late Neolithic I1–Z284 conditional fits, with mode ± 1.96 inverse-Hessian standard deviation. Grey identifies limited constituent-category support. The geographical restrictions overlap dependence blocks and are not independent regional replications

a Shared local recording support



b Whole units and category support

Category	Original	Restricted
L22	236/47	211/46
Z58	206/35	154/33
Other I1	95/20	81/18
Z284	154/21	108/19
U106	157/37	131/33
P312	144/23	123/21
N	125/22	89/20
Other	218/41	189/40

Cells: units / proxy dependence blocks.
1,086 units retained; 249 omitted.
Every outcome category passes the fixed ten-unit / three-block support gate.

Used by a retained unit Other represented position

Every reported locality must have a site within 75 km in ≥ half the draws in both periods.
Points show distinct reported paternal origins; all alternative origins stay together within a unit.

Fig. S2 Geography of the fixed shared-support restriction. Points are distinct positions used by the common-period cohort. Teal marks positions used by at least one retained unit; orange marks other represented positions. A position can belong to more than one unit. The restriction keeps whole units only when every reported locality has recorded-site opportunity in both periods. Category counts report units/proxy dependence blocks before and after restriction. All measurements, reference centres and contrasts remain fixed. National outlines: Natural Earth (undated)

The shared-support comparison retains a unit only when each of its alternative localities has at least one recorded site within 75 km in at least half of the 64 paired realizations in both the Late Neolithic and Roman windows. The criterion uses recorded-site opportunity, not lineage coefficients. It retains 1,086 units and omits 249; all eight categories retain at least ten units and three blocks. The four designs are subsets of the original design matrices, with no refitted component, reference centre, nuisance transformation or composite weight. Each grid's original 399 block/cell weight streams is restricted to retained units and renormalized. These are comparisons in a narrower geographical population, not matched ancient samples or an adjustment for all recording or environmental bias.

Table S17 Common-support conditional contrasts on the frozen measurement

Specification	Late Neolithic I1–Z284	Late Neolithic–Roman difference	Crossed range for difference
Linear / original	0.448 [0.063, 0.833]	0.623 [−0.017, 1.263]	0.565 [−0.630, 1.723]
Linear / shifted	0.565 [0.202, 0.928]	0.601 [−0.034, 1.235]	0.603 [−0.612, 1.637]
Quadratic / original	0.457 [0.068, 0.847]	0.620 [−0.019, 1.260]	—
Quadratic / shifted	0.559 [0.182, 0.937]	0.589 [−0.050, 1.228]	—

All fits retain 1,086 units. The first two numerical columns show conditional mode and inverse-Hessian 95% interval. The last shows median and empirical 2.5–97.5% range over 399 crossed block/cell weights per linear grid; — means no quadratic crossed-weight batch was specified. The archaeological basis, reference centres and composite weights remain fixed.

Current-profile sensitivity to branch representation

The extension uses the same two fixed upward cuts described for Table S13, now projecting all recovered positions through the common four-period and later-profile transformations. Every outcome category is preserved. For each cap, a full-cut cohort is accompanied by a cohort restricted to baseline source-record memberships and their localities, without reselecting tree leaves; the latter retains the full-cut dependence blocks. Whole-unit terrain exclusions are applied separately, so the restricted cohort can contain more units than the full-cut cohort. These comparisons separate the effects of grouping branches from those of adding origins as far as the mappings allow; the represented individuals need not be identical under every grouping.

Restricted-cohort curated_record_ids describe only current retained membership. The original full-cut curated list remains in full_cut_curated_record_ids, so source provenance is preserved separately from the restricted unit. Neither curated-list field enters the likelihood or its controls.

Four common geographical specifications and four later profiles in each cap/cohort combination yield 32 conditional fits. Both common linear grids and the preferred Viking and medieval profiles in each full-cut cohort yield eight posterior jobs. Each cap also receives 400 crossed block/cell reweightings for the original-grid common, preferred Viking and preferred medieval fits (2,400 fits). At least ten units and three proxy dependence blocks are required in every outcome category; all cohorts pass. New cell centres use equally weighted screened source-record localities fixed before estimating the lineage associations, while existing centres, archaeological transforms, nuisance scales, quadratic-coordinate transforms and the original later registry-density reference remain fixed. More source records do not imply more independent families.

Table S18 Current-profile cohorts under both fixed branch cuts

Cap / cohort / scope	Units	Records	Locality rows	Positions	Blocks	Recovered	Lost
2 / Full / Common	1014	2130	2052	1756	202	495	6
2 / Full / Later	1020	2147	2067	1767	202	500	0
2 / Shared / Common	1017	1641	1593	1399	202	0	0
2 / Shared / Later	1020	1647	1598	1404	202	0	0
5 / Full / Common	633	2393	2291	1941	133	768	16
5 / Full / Later	639	2422	2317	1958	133	775	0
5 / Shared / Common	637	1641	1590	1399	133	0	0
5 / Shared / Later	639	1647	1595	1404	133	0	0

Full uses every eligible record under the indicated cut. Shared retains only baseline source-record memberships within the full-cut units and dependence blocks. Recovered/lost counts are relative to the baseline cohort for that scope. Separate whole-unit terrain exclusions explain why a shared cohort can contain more units than its full cohort. All eight outcome categories meet ten-unit/three-block support; the selection inventory retains every category count, block count and membership.

Table S19 Focal posterior contrasts and dependence ranges at both branch cuts

Cap	Model	Contrast	Posterior mean [95% interval]	Crossed median [95% range]
2	Common / original	LN I1–Z284	0.443 [−0.026, 0.917]	0.432 [−0.296, 1.145]
2	Common / original	LN–Roman I1–Z284	0.871 [0.128, 1.609]	0.828 [−0.374, 2.107]

Cap	Model	Contrast	Posterior mean [95% interval]	Crossed median [95% range]
2	Common / shifted	LN I1–Z284	0.492 [0.066, 0.920]	—
2	Common / shifted	LN–Roman I1–Z284	0.947 [0.254, 1.668]	—
2	Viking / preferred	L22–Z58	0.456 [0.071, 0.849]	0.334 [−0.377, 1.071]
2	Viking / preferred	U106–P312	−0.308 [−0.760, 0.132]	−0.216 [−0.918, 0.368]
2	Medieval / preferred	Z284–U106	−0.494 [−0.900, −0.113]	−0.508 [−1.228, −0.001]
2	Medieval / preferred	U106–P312	0.334 [−0.053, 0.738]	0.319 [−0.197, 0.951]
5	Common / original	LN I1–Z284	0.668 [0.003, 1.349]	0.687 [−0.196, 1.561]
5	Common / original	LN–Roman I1–Z284	1.296 [0.231, 2.412]	1.175 [−0.202, 2.641]
5	Common / shifted	LN I1–Z284	0.640 [0.009, 1.299]	—
5	Common / shifted	LN–Roman I1–Z284	1.705 [0.631, 2.865]	—
5	Viking / preferred	L22–Z58	0.634 [0.137, 1.147]	0.365 [−0.769, 1.333]
5	Viking / preferred	U106–P312	−0.163 [−0.756, 0.449]	−0.057 [−0.699, 0.681]
5	Medieval / preferred	Z284–U106	−0.299 [−0.827, 0.219]	−0.372 [−1.295, 0.291]
5	Medieval / preferred	U106–P312	0.218 [−0.311, 0.723]	0.162 [−0.388, 0.902]

Posterior intervals are equal-tail intervals conditional on the specified cohort, measurement and model. Crossed-weight ranges are empirical 2.5–97.5% ranges across 400 fits and are not calibrated confidence or posterior intervals. — denotes the shifted common grid, for which no crossed-weight extension was specified. Every coefficient and all defined contrasts are retained, not only these focal comparisons.

All eight posterior fits in the branch-representation analysis meet the prespecified diagnostic criteria. Across parameters and contrasts, maximum R-hat is 1.0080, and minimum bulk and tail effective sample sizes are 602 and 893. There are no divergences or tree-depth hits. All 3,234 extension conditional fits pass finite-value, gradient and curvature checks.

Table S20 Aggregation and recovered-origin comparisons for the focal current profiles

Cap	Model and contrast	Full-cut mode [95% interval]	Shared-record mode [95% interval]
2	Common / original LN I1–Z284	0.396 [−0.061, 0.854]	0.507 [0.061, 0.953]
2	Common / original LN–Roman I1–Z284	0.789 [0.050, 1.528]	0.860 [0.137, 1.583]
2	Common / shifted LN I1–Z284	0.467 [0.053, 0.881]	0.544 [0.139, 0.949]
2	Common / shifted LN–Roman I1–Z284	0.909 [0.185, 1.633]	0.855 [0.156, 1.554]
2	Viking / preferred L22–Z58	0.435 [0.053, 0.817]	0.365 [0.004, 0.726]
2	Viking / preferred U106–P312	−0.294 [−0.741, 0.153]	−0.383 [−0.821, 0.056]
2	Medieval / preferred Z284–U106	−0.482 [−0.865, −0.099]	−0.480 [−0.847, −0.112]
2	Medieval / preferred U106–P312	0.321 [−0.061, 0.703]	0.358 [−0.010, 0.725]
5	Common / original LN I1–Z284	0.632 [−0.014, 1.278]	0.785 [0.155, 1.416]
5	Common / original LN–Roman I1–Z284	1.186 [0.125, 2.248]	1.253 [0.165, 2.342]
5	Common / shifted LN I1–Z284	0.629 [0.027, 1.231]	0.651 [0.098, 1.203]

Cap	Model and contrast	Full-cut mode [95% interval]	Shared-record mode [95% interval]
5	Common / shifted LN–Roman I1–Z284	1.575 [0.488, 2.661]	1.277 [0.250, 2.305]
5	Viking / preferred L22–Z58	0.595 [0.100, 1.091]	0.471 [0.005, 0.936]
5	Viking / preferred U106–P312	−0.148 [−0.747, 0.450]	−0.314 [−0.906, 0.278]
5	Medieval / preferred Z284–U106	−0.296 [−0.807, 0.214]	−0.413 [−0.888, 0.062]
5	Medieval / preferred U106–P312	0.202 [−0.289, 0.694]	0.272 [−0.199, 0.742]

Intervals use local inverse-Hessian curvature. The full and restricted cohorts use the same branch-cut blocks and fixed transformations; restricting records to baseline membership does not select tree leaves again. These are sensitivity comparisons, not independent replications. The full set of 32 designs also includes both quadratic common-period specifications and both later companion profiles for each branch cap and cohort.

Current-estimator evaluation

The eight simulation conditions were specified after reviewing the empirical profiles and earlier benchmarks, but before generating outcomes. The evaluation uses the current common four-period, three-component design under the original grid. Each condition has 200 datasets and three methods: the unit-level locality-probability mixture, unit-mean covariates, and a locality mixture weighted by the number of represented source records. All use identical priors and controls. Clean conditions collapse alternative localities to their unit means and set multiplicities to one; the three methods are then algebraically identical, so a single fit is reused with that equivalence explicitly recorded.

Category intercepts use smoothed empirical proportions; nonarchaeological coefficients are small seeded synthetic values. Archaeological coefficients are zero except the specified I1–Z284 Component 3 contrast, set to +0.2 in the Late Neolithic and −0.2 in Roman for effect conditions, giving a direct difference of 0.4. Null conditions set both to zero. Successive conditions introduce observed alternative localities, observed source multiplicities, and category-specific block intercepts with standard deviation 0.5. One outcome is drawn per unit from the arithmetic mean of its locality probabilities. A repeated source record repeats that outcome and is not a new independent draw. Blocks and covariates are held fixed when generating an independent test outcome, so prediction concerns the same design and block environment, rather than prediction in new regions.

Every declared dataset and method record is retained, including numerical failures. Measures are conditional-mode bias, root mean square error, inverse-Hessian 95% coverage, mean width, null interval exclusion, and paired expected and sampled log/Brier scores. Monte Carlo uncertainty accompanies performance summaries and paired method differences. This evaluation tests the specified ways of representing records and origins. It does not assess posterior calibration, discovery-stage selection or the historical interpretation of archaeological associations. The separate shared-assay and duplicate-assertion checks test measurement contracts rather than adding fitted replications.

Table S21 Current-estimator performance for the Late Neolithic contrast

Condition	Method	Bias (MCSE)	RMSE	Coverage % (MCSE)	Mean width	Exclude zero %
Clean / null	Mixture	0.017 (0.012)	0.168	96.0 (1.4)	0.678	4.0
Clean / null	Mean covariates	0.017 (0.012)	0.168	96.0 (1.4)	0.678	4.0
Clean / null	Record weight	0.017 (0.012)	0.168	96.0 (1.4)	0.678	4.0
Clean / effect	Mixture	−0.011 (0.012)	0.176	93.5 (1.7)	0.678	17.5

Condition	Method	Bias (MCSE)	RMSE	Coverage % (MCSE)	Mean width	Exclude zero %
Clean / effect	Mean covariates	−0.011 (0.012)	0.176	93.5 (1.7)	0.678	17.5
Clean / effect	Record weight	−0.011 (0.012)	0.176	93.5 (1.7)	0.678	17.5
Origins / null	Mixture	−0.014 (0.012)	0.164	96.0 (1.4)	0.681	4.0
Origins / null	Mean covariates	−0.015 (0.011)	0.163	96.5 (1.3)	0.680	3.5
Origins / null	Record weight	−0.014 (0.012)	0.164	96.0 (1.4)	0.681	4.0
Origins / effect	Mixture	0.006 (0.012)	0.165	96.0 (1.4)	0.681	20.0
Origins / effect	Mean covariates	0.006 (0.012)	0.164	96.5 (1.3)	0.679	19.0
Origins / effect	Record weight	0.006 (0.012)	0.165	96.0 (1.4)	0.681	20.0
Multiplicity / null	Mixture	0.010 (0.012)	0.163	97.5 (1.1)	0.680	2.5
Multiplicity / null	Mean covariates	0.010 (0.011)	0.161	97.5 (1.1)	0.678	2.5
Multiplicity / null	Record weight	0.008 (0.013)	0.180	93.5 (1.7)	0.644	6.5
Multiplicity / effect	Mixture	0.009 (0.011)	0.149	98.5 (0.9)	0.676	18.5
Multiplicity / effect	Mean covariates	0.009 (0.011)	0.149	98.0 (1.0)	0.675	19.0
Multiplicity / effect	Record weight	0.019 (0.011)	0.163	95.5 (1.5)	0.641	29.0
Dependence / null	Mixture	−0.006 (0.012)	0.171	94.0 (1.7)	0.682	6.0
Dependence / null	Mean covariates	−0.006 (0.012)	0.171	94.0 (1.7)	0.681	6.0
Dependence / null	Record weight	−0.007 (0.013)	0.185	94.0 (1.7)	0.648	6.0
Dependence / effect	Mixture	0.020 (0.011)	0.151	97.5 (1.1)	0.676	21.0
Dependence / effect	Mean covariates	0.020 (0.011)	0.152	97.0 (1.2)	0.675	21.5
Dependence / effect	Record weight	0.022 (0.012)	0.164	95.5 (1.5)	0.641	25.0

Each condition contains 200 datasets. Parentheses give Monte Carlo standard errors, on the displayed scale. Coverage and exclusion concern mode ± 1.959964 inverse-Hessian standard deviations, not posterior intervals. The simulated effect is 0.2 for the Late Neolithic contrast and 0.4 for its direct difference from Roman; both contrasts are zero under the null. Under the null, an interval excluding zero is a false exclusion; under the effect condition, it indicates detection. All conditions and methods are retained.

Table S22 Current-estimator performance for the direct Late Neolithic–Roman difference

Condition	Method	Bias (MCSE)	RMSE	Coverage % (MCSE)	Mean width	Exclude zero %
Clean / null	Mixture	0.032 (0.021)	0.295	95.5 (1.5)	1.130	4.5
Clean / null	Mean covariates	0.032 (0.021)	0.295	95.5 (1.5)	1.130	4.5
Clean / null	Record weight	0.032 (0.021)	0.295	95.5 (1.5)	1.130	4.5
Clean / effect	Mixture	−0.050 (0.021)	0.295	95.0 (1.5)	1.133	21.0
Clean / effect	Mean covariates	−0.050 (0.021)	0.295	95.0 (1.5)	1.133	21.0
Clean / effect	Record weight	−0.050 (0.021)	0.295	95.0 (1.5)	1.133	21.0
Origins / null	Mixture	−0.016 (0.019)	0.275	97.0 (1.2)	1.136	3.0
Origins / null	Mean covariates	−0.017 (0.020)	0.276	97.5 (1.1)	1.134	2.5
Origins / null	Record weight	−0.016 (0.019)	0.275	97.0 (1.2)	1.136	3.0
Origins / effect	Mixture	−0.019 (0.021)	0.297	93.0 (1.8)	1.138	27.5
Origins / effect	Mean covariates	−0.020 (0.021)	0.298	93.0 (1.8)	1.136	26.5
Origins / effect	Record weight	−0.019 (0.021)	0.297	93.0 (1.8)	1.138	27.5
Multiplicity / null	Mixture	0.009 (0.019)	0.267	95.5 (1.5)	1.132	4.5
Multiplicity / null	Mean covariates	0.009 (0.019)	0.263	97.0 (1.2)	1.129	3.0
Multiplicity / null	Record weight	0.008 (0.020)	0.277	94.0 (1.7)	1.080	6.0
Multiplicity / effect	Mixture	−0.014 (0.019)	0.264	96.5 (1.3)	1.133	24.5

Condition	Method	Bias (MCSE)	RMSE	Coverage % (MCSE)	Mean width	Exclude zero %
Multiplicity / effect	Mean covariates	−0.013 (0.019)	0.263	96.0 (1.4)	1.130	25.5
Multiplicity / effect	Record weight	−0.010 (0.020)	0.280	95.0 (1.5)	1.081	29.5
Dependence / null	Mixture	−0.029 (0.019)	0.275	96.5 (1.3)	1.138	3.5
Dependence / null	Mean covariates	−0.030 (0.019)	0.275	96.5 (1.3)	1.136	3.5
Dependence / null	Record weight	−0.033 (0.020)	0.291	94.0 (1.7)	1.089	6.0
Dependence / effect	Mixture	0.008 (0.017)	0.242	99.5 (0.5)	1.132	25.5
Dependence / effect	Mean covariates	0.006 (0.017)	0.241	98.5 (0.9)	1.130	26.0
Dependence / effect	Record weight	0.017 (0.019)	0.264	95.5 (1.5)	1.084	29.0

Each condition contains 200 datasets. Parentheses give Monte Carlo standard errors, on the displayed scale. Coverage and exclusion concern mode ± 1.959964 inverse-Hessian standard deviations, not posterior intervals. The simulated effect is 0.2 for the Late Neolithic contrast and 0.4 for its direct difference from Roman; both contrasts are zero under the null. Under the null, an interval excluding zero is a false exclusion; under the effect condition, it indicates detection. All conditions and methods are retained.

Table S23 Paired predictive performance of the current unit-level mixture

Condition	Log gain vs mean covariates	Brier gain vs mean covariates	Log gain vs record weight	Brier gain vs record weight
Clean / null	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Clean / effect	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Origins / null	−0.423 (0.081)	−0.025 (0.017)	0.000 (0.000)	0.000 (0.000)
Origins / effect	−0.432 (0.084)	−0.042 (0.017)	0.000 (0.000)	0.000 (0.000)
Multiplicity / null	−0.486 (0.089)	−0.043 (0.017)	14.440 (0.283)	3.559 (0.079)
Multiplicity / effect	−0.375 (0.081)	−0.012 (0.016)	13.902 (0.263)	3.418 (0.074)
Dependence / null	−0.467 (0.089)	−0.044 (0.017)	13.930 (0.271)	3.464 (0.074)
Dependence / effect	−0.338 (0.080)	−0.010 (0.016)	13.418 (0.261)	3.267 (0.068)

Values and Monte Carlo standard errors in parentheses are multiplied by 1,000. Positive gains favour the probability mixture: log gain is mixture minus comparator; Brier gain is comparator minus mixture. Natural-log scores and eight-category Brier loss are averaged over units, without division of Brier loss by eight. Expected scores integrate over the generating category probabilities at the same covariates and block effects; sampled independent test-label scores are also retained. These are not predictions in unseen regions.

The experiment produced 4,800 method records with no numerical failures. Reusing fitted solutions for algebraically identical methods reduced the number of optimizations to 3,600. Their median recorded elapsed time is 3.17 s (10th–90th percentiles 2.72–3.70 s) in the checked environment. These timings include model fitting and score calculation within each method and reflect concurrent execution; they are descriptive implementation costs, not hardware-independent speed claims.

Transfer to another archaeological setting

A new application begins with the historical alternatives and the observations capable of distinguishing them. Define the meaning of a paternal origin, the branch nomenclature and unit reconciliation rules before interpreting archaeological proximity. Define the archaeological assertion and retain original wording, location, material, context, date and phase relationships. A feature dictionary should distinguish activities the sources can support, without making a reported feature a social or ethnic identity.

Fix a compact set of periods and features that address the question, then assess count and geographical support before estimating lineage associations. Specify the kernel, positional conventions, recording and opportunity measures. A common basis is appropriate when the same archaeological distinction can be

supported across periods. Where activities or sources require different measurements, retain them as separate comparisons and document their historical connection. Preserve the complete transform, including centring, scaling, orientation and any rotation, so a reported coefficient can be reconstructed.

Choose controls in relation to the historical and observation processes. For example, a karst-based application may require controls for geology or interactions between geology and archaeological proximity. Swedish terrain controls and distance scales are not universal defaults. Justify the variables and scales for the new setting, and keep them fixed across comparisons intended to isolate a change in period or source pool.

For inference, preserve the chosen unit-level likelihood, contrast definitions and prior scale across matched sensitivities. Distinguish a change in measurement from a change in represented units or statistical adjustment. Choose a finite set of checks for the identified sources of dependence and geographical or source uncertainty, with their scope fixed independently of the resulting coefficients. Record which results were already known and retain relevant comparisons with insufficient support or unfavourable results. Historical interpretation then states which setting is associated, which period differences are directly estimated, and what independently dated evidence would distinguish persistence from later sorting.

The reusable numerical functions accept supplied sources, arrays and frozen settings rather than assuming Swedish paths. A compact worked example can make the steps from source records to archaeological measurements and lineage contrasts easier to inspect before scaling up to a national dataset. Reproduction rebuilds those measurements and model results from their scientific inputs; stored coefficient tables allow the outputs to be inspected. Source access, redistribution rights and living-person information remain distinct from numerical reproducibility.

The runnable synthetic example contains shared laboratory dates, two archaeological periods, a fixed building-minus-hearth axis and one likelihood contribution per paternal unit. An optional all-multiple-origin mode tests the same interface with no singleton units. Contract checks cover empty and singleton feature panels, identical date allocation for shared determinations, duplicate-assertion invariance, exact joint-site loss, and compiled Stan/Python likelihood and gradient agreement. Zero exposure applies only within the supplied observed universe; insufficient or unknown archaeological coverage remains a separate support decision. The software and reconstruction instructions are identified by https://github.com/nstanisic1/SCLARA and https://doi.org/10.5281/zenodo.24088352 (version 2026.11.09.4).

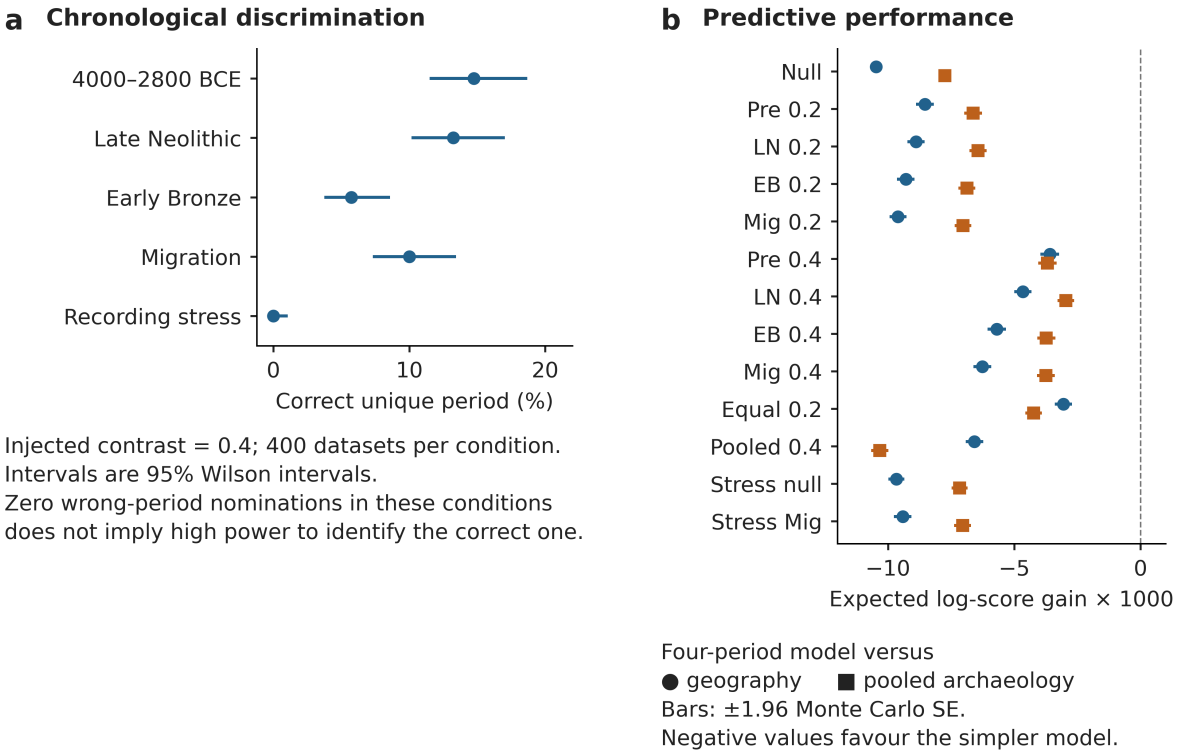


Fig. S3 Chronological and predictive benchmarks for the earlier posthole–hearth measurement. a Correct unique-period nomination in all four single-period 0.4 conditions and the selective-recording Migration condition, with 95% Wilson intervals over 400 datasets each. b Paired expected log-score differences of the four-period model against geography-only and pooled-archaeology comparators across all thirteen conditions, multiplied by 1,000; bars show ±1.96 Monte Carlo standard errors. Negative differences favour the simpler comparator. Pre denotes 4000–2800 BCE, LN Late Neolithic, EB Early Bronze and Mig Migration. Complete conditions, Brier scores and interpretation are retained in Tables S11–S12

Complete component comparisons and allocation sensitivity

Online Resource 4 gives all 840 retained common-period estimates for the three components and seven lineage contrasts. Figure S4 places the I1–Z284 comparison alongside the other components. Table S24 summarizes the separate fits to 64 paired material-date and archaeological-position allocations under each linear grid.

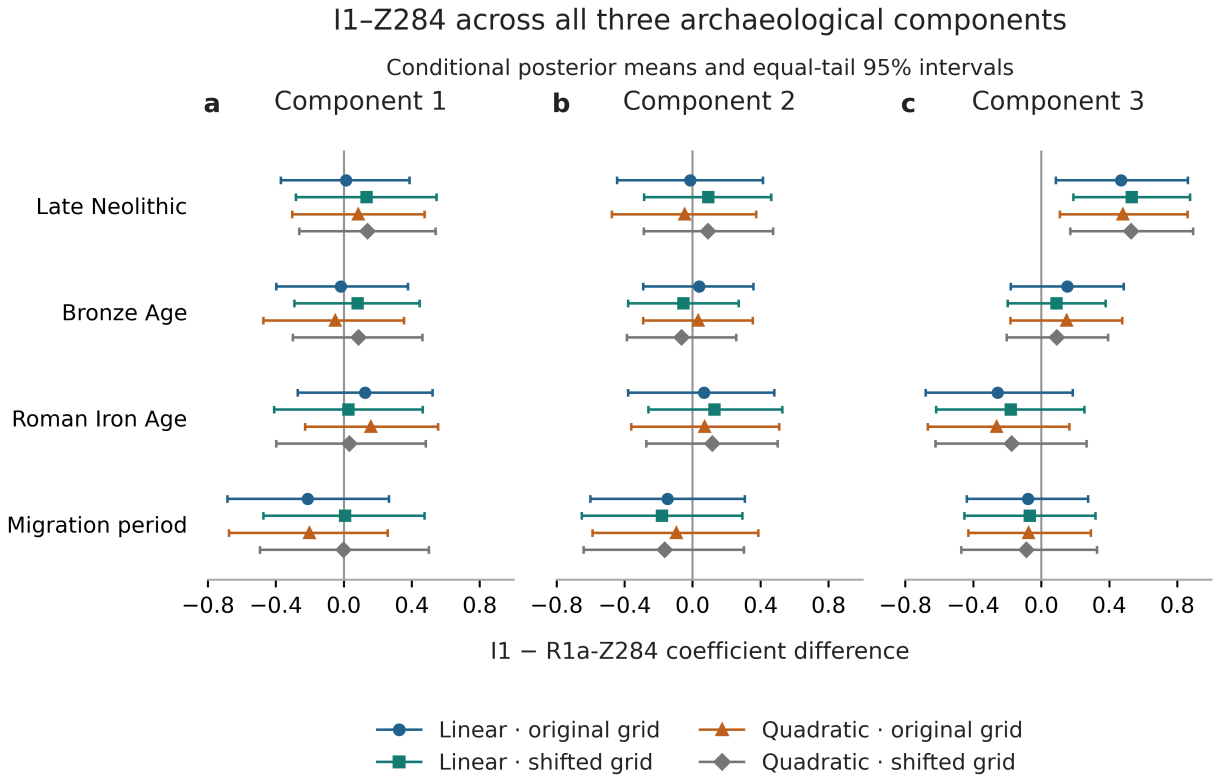


Fig. S4 Conditional posterior means and equal-tail 95% intervals for I1–R1a-Z284 in each period, shown for all three common archaeological components and all four national model/grid specifications. Positive values indicate a larger archaeological slope for composite I1, which retains the fixed 237:207:95 category weights. The original and shifted grids differ by 100 km. All specifications use the same observations; they describe model/grid sensitivity rather than independent replications. Intervals are conditional on the stated exploratory comparisons and do not adjust for the preceding search. Direct period differences are reported in the complete comparison table. The shared horizontal scale displays coefficients for distinct archaeological components; their meanings remain defined by their respective fixed loadings

Table S24 Current common-period allocation sensitivity

Contrast	Grid	Median	Empirical 95% range	Mode sign	95% interval exclusion
Late Neolithic I1–Z284	Original	0.369	[0.254, 0.559]	64/64 positive	39/64
Late Neolithic–Roman I1–Z284 difference	Original	0.558	[0.350, 0.883]	64/64 positive	35/64
Late Neolithic Z284–U106	Original	−0.434	[−0.623, −0.278]	64/64 negative	43/64
Late Neolithic I1–Z284	100 km shifted	0.430	[0.284, 0.586]	64/64 positive	59/64
Late Neolithic–Roman I1–Z284 difference	100 km shifted	0.594	[0.297, 0.819]	64/64 positive	38/64
Late Neolithic Z284–U106	100 km shifted	−0.413	[−0.585, −0.280]	64/64 negative	41/64

Summary of the 64 paired material-date and archaeological-position allocations fitted under each linear grid specification. All measurements retain the common seven-family definitions and fixed reference transformation. Empirical ranges summarize variation among conditional modes; they are neither posterior nor calibrated confidence intervals. The last column counts allocations whose mode ± 1.959964 inverse-Hessian standard error excludes zero in the reported direction. Counts describe these retained allocations and should not be read as posterior probabilities or independent replications. These results concern the current seven-family measurement, not the earlier posthole–hearth source-pool analysis in Table S9.

References

Allentoft ME, Sikora M, Fischer A et al. (2024a) 100 ancient genomes show repeated population turnovers in Neolithic Denmark. Nature 625:329–337. https://doi.org/10.1038/s41586-023-06862-3

Allentoft ME, Sikora M, Refoyo-Martínez A et al. (2024b) Population genomics of post-glacial western Eurasia. Nature 625:301–311. https://doi.org/10.1038/s41586-023-06865-0

Altner A (2014) Planerade vägförstärkningar av väg 708, Bobygden. Rapport Länsmuseet Gävleborg 2014:21. https://lansmuseetgavleborg.se/wp-content/uploads/2021/02/xlm_rapport_2014-21.pdf

Altner A (2016) Förhistorisk kolning i Färila. Arkeologisk utredning och undersökning. Rapport Länsmuseet Gävleborg 2016:12. https://pub.raa.se/dokumentation/b4552c3f-27e6-4416-b778-9f591e09f059/original

Andersson T (2004) Odlings- och boplatslämningar från järnålder. Område 3, Mölletofta–Rya, E4-projektet i norra Skåne. UV Syd, Dokumentation av fältarbetsfasen 2004:21. Riksantikvarieämbetet. https://pub.raa.se/dokumentation/070019c0-94b0-40cb-b740-eda7b8491829/original

Balknäs N (with contributions by Karlenby L) (2020) En järnåldersgård i Marieberg. Arkeologgruppen rapport 2020:21. https://arkeologgruppen.se/wp-content/uploads/2020/08/2018_64-Marieberg-KLAR-o-GODK_web-pdf-A-1.pdf

Battey CJ, Ralph PL, Kern AD (2020) Space is the Place: Effects of Continuous Spatial Structure on Analysis of Population Genetic Data. Genetics 215:193–214. https://doi.org/10.1534/genetics.120.303143

Bolander A (2016) Fjärrvärmeledning mellan Örtofta och Landskrona. Arkeologerna rapport 2016:33. https://pub.raa.se/dokumentation/276a8e51-dd32-44d0-9d91-04a9d4a9e36a/original

Boström LA (undated) Y-DNA-haplogrupper i södra Norrbotten. Source compilation, snapshot 3 November 2026. https://familjenbostrom.se/genealogi/dna/nobo/

Carlie A (2002) Människor och landskap. Om förhistoriska samhällen i en nordskånsk inlandsbygd. In: Carlie A (ed) Skånska regioner. Tusen år av kultur och samhälle i förändring. Riksantikvarieämbetet, Arkeologiska undersökningar, Skrifter 40. Riksantikvarieämbetets förlag, Stockholm, pp. 280–361. https://pub.raa.se/dokumentation/34ef1548-eba8-4796-8e77-1fb0c4556b8c/original/1

Emilsson A, Lundholm S, Vestbö Franzén Å (2020) Öjabymotet. Steg 2 utredning och förundersökning 2019. Kalmar läns museum, arkeologisk rapport 2020:01. https://kulturparkenberattar.se/Rapporter%20officiella/2020.1_KLM_Ojabymotet_AU2_och_FU.pdf

Ericson T (2007) Utbyggnadsområde 130 Mårtens Fälad. Boplatslämningar från brons- och järnålder. UV Syd rapport 2007:15. Riksantikvarieämbetet. https://pub.raa.se/dokumentation/3777d1a7-083a-48c4-9180-f6f4db709bb0/original

Ericsson A, Franzén G (2005) Hägnadsmurar och rydskogar. Stensträngsområdet söder om Väderstads samhälle delområde 1, 2, 3, 4 & 6. UV Öst rapport 2005:70. Riksantikvarieämbetet. https://pub.raa.se/dokumentation/266d2299-9f98-4956-a25e-061ee66b4b42/original

Eurostat (2024) GISCO territorial units for statistics (NUTS), 2024 edition, level 2, 1:1 million, EPSG:4326. https://gisco-services.ec.europa.eu/distribution/v2/nuts/

Fagerlund D (2016) För bröder, präster och kungar. En medeltida tegelindustri i Uppsala. Upplandsmuseets rapporter 2016:03. https://www.upplandsmuseet.se/globalassets/publikationer/rapportserien/rapporter-2016/rapport-2016_03-akademiska-sjukhusomradet.pdf

FamilyTreeDNA (2026) Sweden DNA project and public Y-DNA haplotree. Source snapshots 2 November 2026. https://www.familytreedna.com/public/Sweden?iframe=ydna-results-overview

Färjare A (2021) Röjningar i Rödje. Arkeologisk förundersökning av del av röjningsröseområde inom Rödjes gamla tomts ägor. Eldrun arkeologiska skrifter 2021:7. Eldrun kulturmiljö AB. https://pub.raa.se/dokumentation/a56edeac-96af-4c26-bed0-467066992bbe/original

GeoNames (2026) GeoNames geographical database, Sweden export (SE.zip). Snapshot November 2026. https://download.geonames.org/export/dump/SE.zip

Gneiting T, Raftery AE (2007) Strictly proper scoring rules, prediction, and estimation. Journal of the American Statistical Association 102:359–378. https://doi.org/10.1198/016214506000001437

Helander C (2013) Kvarnbacken. UV Rapport 2013:88. Riksantikvarieämbetet. https://pub.raa.se/dokumentation/531a3159-530d-4e13-a9a4-fa6b7a787f54/original

Hellgren M (2023) Röjningsrösen i Kråkhult. Arkeologisk förundersökning. Göta Arkeologi rapport 2023:10. https://files.builder.misssite.com/b9/9b/b99b7a16-f2d7-4bca-a589-c1655779deb8.pdf

Hennius A (2020) Towards a refined chronology of prehistoric pitfall hunting in Sweden. European Journal of Archaeology 23:530–546. https://doi.org/10.1017/eaa.2020.8

Hennius A (2022) Compilation of 14C-dated trapping pits in 2020, version 1. Swedish National Data Service, Uppsala University. https://doi.org/10.57804/vn18-6x14

Henriksson C (2023) Arkeologisk förundersökning av fornlämningarna L1962:9924, L2021:3006, L2021:3007, L2021:3014, L2021:3017, L2021:3022 och L2021:3024 samt fastställande av status för lämningarna L1962:9925, L1962:9975 och L2021:3015, Mariestads kommun. Västergötlands museum rapport 2023:1. https://pub.raa.se/dokumentation/075ecc8f-aabb-4264-a3d8-3c204a111962/original

Holmqvist S (2021) Kolningsgrop och tjärdal i Leksand. Rapporter från Arkeologikonsult 2021:3435. https://www.arkeologikonsult.se/rapporter/2021.3435%20Kolning%20och%20tj%C3%A4rdal%20i%20Leksand.pdf

Hothorn T, Bretz F, Westfall P (2008) Simultaneous inference in general parametric models. Biometrical Journal 50:346–363. https://doi.org/10.1002/bimj.200810425

HydroSHEDS (undated) HydroSHEDS version 1 elevation data, Europe 30 arcsecond grid. Exact crop and original-file hash retained in the source archive. https://www.hydrosheds.org/products/hydrosheds

Institute for Language and Folklore (2026) Place Name Service: parish and province references. Snapshot November 2026. https://placenameservice.isof.se/

Kinnunen J (undated) Fennoscandia – Y-DNA haplogroup I1,I2, N and R1a,R1b. Visual DNA, interactive map and source description. Snapshot 3 November 2026. https://www.visual-dna.net/visualized-data-articles

Kjellberg J (2006) Harbo-Smedsbo. Järnframställning och boplatslämningar i Heby. Kulturmiljövård Mälardalen rapport 2006:1. https://pub.raa.se/dokumentation/6a6569e2-3aa7-4289-9777-47f1652d5580/original

Lageräs P (2002) Skog, slåtter och stenröjning. Paleoekologiska undersökningar i trakten av Stoby i norra Skåne. In: Carlie A (ed) Skånska regioner. Tusen år av kultur och samhälle i förändring. Riksantikvarieämbetet, Arkeologiska undersökningar, Skrifter 40. Riksantikvarieämbetets förlag, Stockholm, pp. 362–411. https://pub.raa.se/dokumentation/34ef1548-eba8-4796-8e77-1fb0c4556b8c/original/1

Lehner B, Verdin K, Jarvis A (2008) New global hydrography derived from spaceborne elevation data. Eos, Transactions American Geophysical Union 89:93–94. https://doi.org/10.1029/2008EO100001

Lindman G (2001) Räppe industriområde. Arkeologisk utredning och förundersökning. UV Väst rapport 2001:4; Smålands museum rapport 2001:1. Riksantikvarieämbetet and Smålands museum. https://kulturparkenberattar.se/Rapporter%20officiella/2001_1_Rappe_ind.omr._Bergunda_RAA50%2C_152_och_158._AU_o_FU.pdf

Lindman G (2003) Räppe industriområde – arkeologisk förundersökning av boplatslämningar i ett röjningsrösområde. Smålands museum rapport 2003:24. https://kulturparkenberattar.se/Rapporter%20officiella/2003_24_Rappe_ind.omr._FU_RAA158_o_50.pdf

Lucas M, Lucas R (2013) Gårdar och hästoffer. Järnålder och tidig medeltid i Fyrislund. Upplandsmuseets rapporter 2013:02. https://www.upplandsmuseet.se/globalassets/publikationer/rapportserien/rapporter-2013/hellby1217_lagupplost.pdf

Malmström H, Günther T, Svensson EM et al. (2019) The genomic ancestry of the Scandinavian Battle Axe Culture people and their relation to the broader Corded Ware horizon. Proceedings of the Royal Society B 286:20191528. https://doi.org/10.1098/rspb.2019.1528

Margaryan A, Lawson DJ, Sikora M et al. (2020) Population genomics of the Viking world. Nature 585:390–396. https://doi.org/10.1038/s41586-020-2688-8

McColl H, Kroonen G, Moreno-Mayar JV et al. (2026) Tracing the spread of Germanic languages using ancient genomics. Nature. https://doi.org/10.1038/s41586-025-09064-1

Morris TP, White IR, Crowther MJ (2019) Using simulation studies to evaluate statistical methods. Statistics in Medicine 38:2074–2102. https://doi.org/10.1002/sim.8086

Natural Earth (undated) Admin 0 – Countries, 1:10 million. Public-domain map data; retained snapshot identified by checksum. https://www.naturalearthdata.com/downloads/10m-cultural-vectors/10m-admin-0-countries/

Nordin P (2023) Sju hus från yngre stenålder vid Mannaberg i Skrea. Arkeologerna rapport 2023:130. Statens historiska museer. https://pub.raa.se/dokumentation/62c34985-614d-48bc-9e7b-ae9297fb59ba/original

Owen AB, Eckles D (2012) Bootstrapping data arrays of arbitrary order. Annals of Applied Statistics 6:895–927. https://doi.org/10.1214/12-AOAS547

Petersson M (2001) Tre åkerundersökningar i Östergötland. Arkeologiska undersökningar av fossil åkermark, gravar och boplatslämningar från äldre järnålder. UV Öst rapport 2001:33. Riksantikvarieämbetet. https://pub.raa.se/dokumentation/248eaca9-7b9a-4741-8981-e66e22c7236f/original

Petersson M (ed) (2015) Farstorp – ett röjningsröseområde i långtidsperspektiv. Arkeologisk undersökning. Rapport 2015:116. Statens historiska museer. https://pub.raa.se/dokumentation/0b0ce9ca-f6b6-460a-ad47-7ff31470e2be/original

Petersson M, Zetterlund P (2021) Glasskärvan 1 boplatslämningar och fossil åker. Arkeologisk förundersökning och undersökning. Arkeologerna rapport 2021:4. Statens historiska museer. https://pub.raa.se/dokumentation/917caba5-3a67-4944-8184-ec5be35925d7/original

Pinotti T, McColl H (2025) Y-chromosome data from “Steppe Ancestry in Western Eurasia and the Spread of the Germanic Languages”, McColl et al. 2025. Zenodo. https://doi.org/10.5281/zenodo.15163112

Ralf A, Montiel González D, Zhong K, Kayser M (2018) Yleaf: Software for Human Y-Chromosomal Haplogroup Inference from Next-Generation Sequencing Data. Molecular Biology and Evolution 35:1291–1294. https://doi.org/10.1093/molbev/msy032

Ralf A, Montiel González D, Zhong K, Kayser M (undated) Yleaf 2.1 hg19 marker dictionary. Revision a367282f5ee621032a3721c73d0bd8b957d9f9b4. https://github.com/genid/Yleaf/blob/a367282f5ee621032a3721c73d0bd8b957d9f9b4/Position_files/hg19.txt

Reimer PJ, Austin WEN, Bard E et al. (2020) The IntCal20 Northern Hemisphere radiocarbon age calibration curve (0–55 cal kBP). Radiocarbon 62:725–757. https://doi.org/10.1017/RDC.2020.41

Riksantikvarieämbetet (2026) Kulturhistoriska lämningar. Kulturmiljöregistret geodata, national snapshot 3 November 2026. https://www.raa.se/hitta-information/oppna-data/oppna-data-portal/

Rimpi A (2025) Hunting pits as temporal manifestations of landscape domestication–large-scale sampling to identify temporality in land use. Journal of Archaeological Method and Theory 32:56. https://doi.org/10.1007/s10816-025-09727-3

Ring C, Pedersen EA, Gurstad-Nilsson H et al. (2001) Kölbygärde. Bronsåldersboplatser i ett fossilt odlingslandskap. E22-projektet rapport 2001:10. Kalmar läns museum. https://kalmarlansmuseum.se/wp-content/uploads/2022/05/kolbygarde.pdf

Rodríguez-Varela R, Moore KHS, Ebenesersdóttir SS et al. (2023) The genetic history of Scandinavia from the Roman Iron Age to the present. Cell 186:32–46.e19. https://doi.org/10.1016/j.cell.2022.11.024

Seiler A, Sjölin M (eds) (2022) Gravar och gårdsmiljöer vid Bådstorp. Arkeologisk undersökning. Arkeologerna rapport 2022:3. Statens historiska museer. https://pub.raa.se/dokumentation/5787a9ae-1cc9-4cd5-b763-0fef8c789979/original

Speidel L, Silva M, Booth T et al. (2025) High-resolution genomic history of early medieval Europe. Nature 637:118–126. https://doi.org/10.1038/s41586-024-08275-2

Strucke U (2026) Strucke C14 Sweden, version 2. Zenodo. https://doi.org/10.5281/zenodo.21932353

Strucke U, Bäck M (2003) Förhistoriska lämningar och förindustriell tegeltillverkning i Selånger. UV Mitt, Dokumentation av fältarbetsfasen 2003:1. Riksantikvarieämbetet. https://pub.raa.se/dokumentation/a9178d81-57db-4d8a-b151-14a7f3bab07b/original

Torstensdotter Åhlin I, Skoglund P, Munkenberg BA, Gustafsson P (1998) Röjningsrösen och gravar i Hamneda socken. Rapport UV Syd 1998:2; Smålands museum rapport 1998:1. Riksantikvarieämbetet and Smålands museum. https://kulturparkenberattar.se/Rapporter%20officiella/1998_1_SU_RAA72_73_77_78_87%2Csamt_250.2_och_250.3_Hamneda_sn.pdf

Vehtari A, Gelman A, Simpson D, Carpenter B, Bürkner PC (2021) Rank-normalization, folding, and localization: an improved R-hat for assessing convergence of MCMC (with Discussion). Bayesian Analysis 16:667–718. https://doi.org/10.1214/20-BA1221

Wennerberg R (2008) 1000 år av kolning i Nifsarp. Jönköpings läns museum, arkeologisk rapport 2008:17. https://jonkopingslansmuseum.se/wp-content/uploads/2018/01/2008_17.pdf

YFull (2024) YFull YTree, version 12.00.0. Released 16 February 2024. Versioned tree archive. https://github.com/YFullTeam/YTree/blob/master/ytree/tree_12.00.0.zip